Design of an Attitude Control System for Earth Observation CubeSats

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by

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Chinese Abstract

本论文提出了 NPUSat，一个假设的地球观测立方星任务，的姿态控制系统（ACS）设计与仿真。该 ACS 采用非线性模型预测控制（MPC）实现纳迪尔指向和目标跟踪模式下的精确指向，同时补偿轨道扰动。仿真基于 Python 实现，利用优化数值库（NumPy 处理数组运算，SciPy 处理科学计算）和专用工具（包括用于控制优化的 do-mpc、用于轨道传播的 SGP4 和用于地磁场建模的 pyIGRF）。系统满足任务要求：保持纳迪尔指向和地面目标跟踪（误差 <1°）。采用四面体构型的冗余反作用飞轮确保容错能力，并通过单执行器故障的应急模式仿真验证。结果凸显了模型预测控制在立方星姿态控制中的有效性，兼具最优性能和抗干扰鲁棒性。

关键词：立方星，近地轨道，姿态控制，模型预测控制，最优控制

Abstract

This thesis presents the design and simulation of an attitude control system (ACS) for NPUSat, a hypothetical CubeSat mission for Earth observation. The ACS employs nonlinear Model Predictive Control (MPC) to achieve precise pointing in Nadir and target tracking modes while compensating for orbital disturbances. Implemented in Python, the simulation leverages optimized numerical libraries (NumPy for array operations, SciPy for scientific computing) and specialized tools including do-mpc for control optimization, SGP4 for orbit propagation, and pyIGRF for geomagnetic field modeling. The system demonstrates compliance with mission requirements: maintaining Nadir pointing, and tracking ground targets (<1° error). Redundant reaction wheels in a tetrahedral configuration ensure fault tolerance, validated through contingency-mode simulations with one failed actuator. The results highlight Model Predictive Control efficacy in attitude control of CubeSats, combining optimal performance with robustness against disturbances.

Keywords: CubeSat, Low Earth Orbit, Attitude Control, Model Predictive Control, Optimal Control.

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Abbreviations

ACS	Attitude Control System
ADS	Attitude Determination System
ADCS	Attitude Determination and Control System
BLDC	Brushless DC Motor
CoM	Center of Mass
CoP	Center of Pressure
EKF	Extended Kalman Filter
GC	Geometric Center
IGRF	International Geomagnetic Reference Field
JD	Julian Date
KF	Kalman filter
LEO	Low Earth Orbit
LEOP	Launch and Early Orbit Phase
MoI	Moment of Inertia
MPC	Model Predictive Control
NORAD	North American Aerospace Defense Command
RAAN	Right Ascension of the Ascending Node
RPM	Rotations Per Minute
RWRF	Reaction Wheel Reference Frame
SDP	Simplified Deep-space Perturbations
SGP	Simplified General Perturbation
TLE	Two-Line orbital Element set
TSI	Total Solar Irradiance
UKF	Unscented Kalman Filter
UT	Universal Time

Notation

In the following, a short description of the notation used in this thesis is given.

Mathematical:

Vectors:	$\vec{\boldsymbol{r}}$
Unit Vectors:	$\hat{\boldsymbol{r}}$
Two-norm:	$\|\vec{\boldsymbol{r}}\|$
Matrices:	$\underline{\mathbf{R}}$
Identity Matrix:	$\underline{\mathbf{I}}$
Inertia Matrix:	$\underline{\mathbf{J}}$
Matrix of ones:	$\underline{\mathbf{1}}$
Matrix of zeros:	$\underline{\mathbf{0}}$
Quaternions:	$\mathbf{q}$

Reference frames:

ECI	Earth-Centered Inertial reference frame:	i
ECEF	Earth-Centered Earth-Fixed reference frame:	e
ORF	Orbit Reference Frame:	o
TRF	Target Reference Frame:	t
BRF	Body Reference Frame:	b
CRF	Control Reference Frame:	c
GRF	Global Reference Frame:	g

A vector defined in some reference frame will be denoted ${}^{i}\vec{\boldsymbol{r}}$ if for example $\vec{\boldsymbol{r}}$ is in the ECI frame. Rotations given by quaternions or matrices are denoted by ${}^{o}_{i}\mathbf{q}$ or ${}^{o}_{i}\underline{\mathbf{R}}$ for a rotation from the ECI frame to the ORF frame.

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Chapter 1

Introduction

The rapid advancements in small satellite technology have revolutionized the aerospace sector, with CubeSats emerging as a dominant platform for space research and exploration. Originating in 1999 as a collaborative project between California Polytechnic State University and Stanford University, CubeSats were designed to reduce development costs, shorten production cycles, and increase accessibility to space [CalPoly, 2022]. With standardized dimensions based on a 10×10×10 cm unit (1U) and masses of up to 2 kg per unit, CubeSats are scalable and versatile, supporting configurations from 1U to 12U.

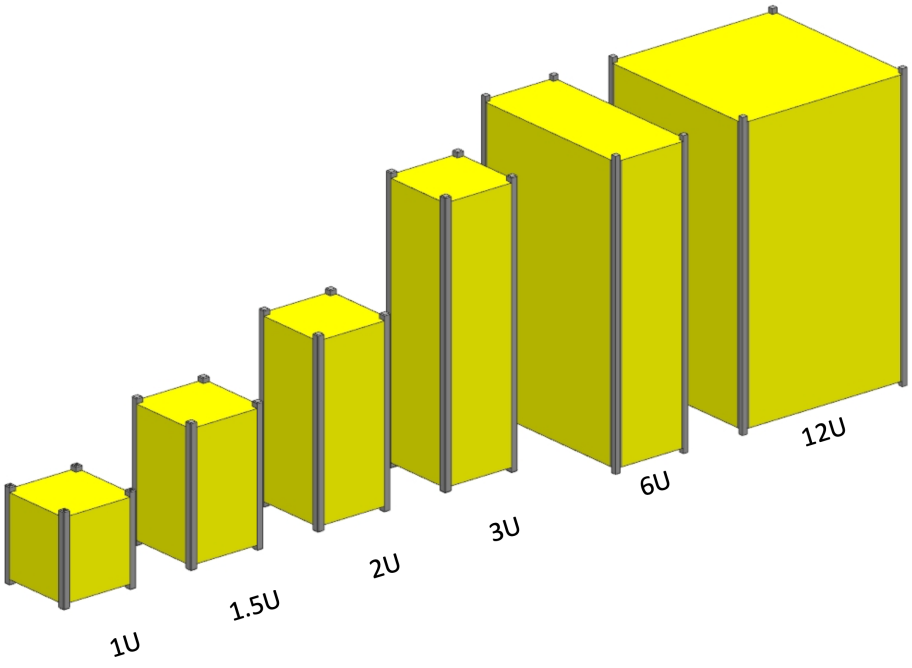


Figure 1.1: The current CubeSat Family (1U-12U).

These miniature satellites are frequently deployed into Low Earth Orbit (LEO) to perform diverse missions, including Earth observation [Praks, 2015], telecommunication [Larsen, 2011], and technology demonstration [Yu, 2016]. Their low cost, modular design, and ability to be launched as secondary or tertiary payloads aboard large rockets have democratized access to space, enabling universities, startups, and small research institutions to increasingly participate in space exploration, with 2596 CubeSats launched as of December 31, 2024 [Kulu, 2025]. Companies like Planet Labs have demonstrated the commercial potential of CubeSats by deploying large constellations, referred to as "Flocks", capable of capturing daily high-resolution imagery of the Earth [Safyan, 2020], solidifying CubeSats as a cornerstone of modern space technology, and expanding Earth observation capabilities.

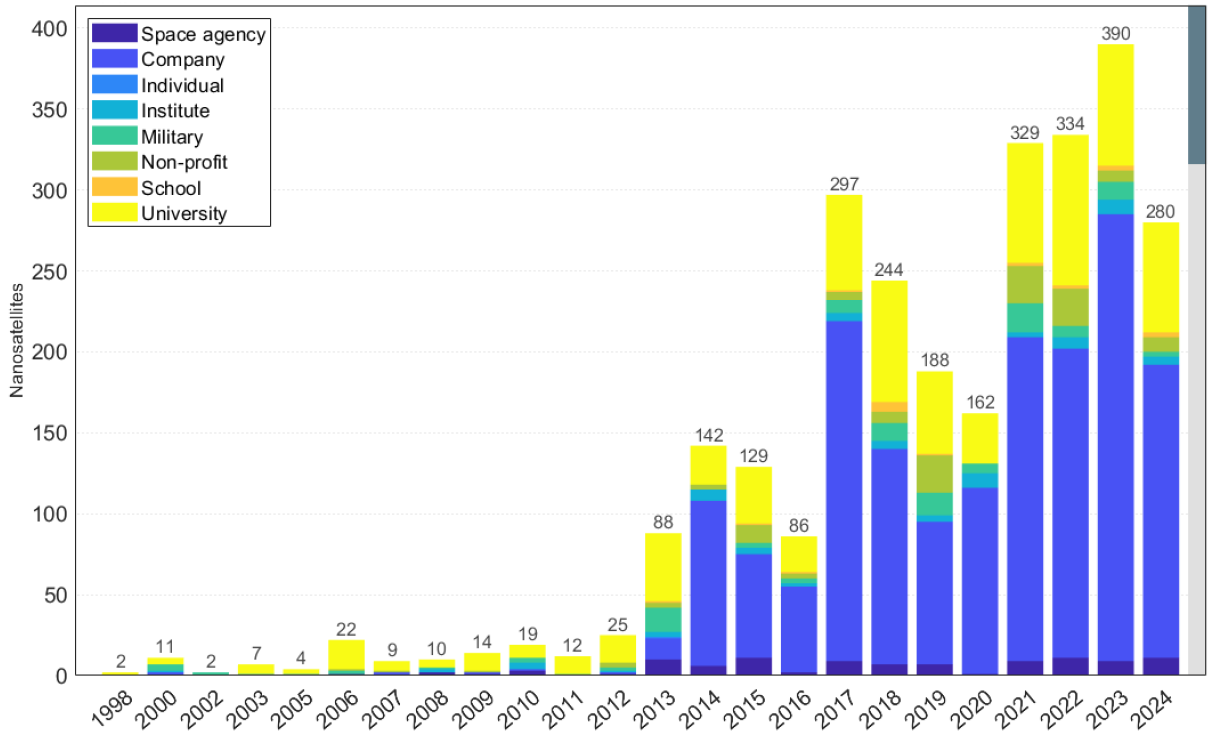


Figure 1.2: Nanosatellite Launches by Organisations (source: nanosat.eu).

For a CubeSat to accomplish its mission, it needs an Attitude Determination and Control System (ADCS), which is a fundamental part of any satellite [Markley & Crassidis, 2014]. The ADCS is responsible for all maneuvers related to the orientation of the satellite in space. This includes detumbling during the early orbit phase, pointing and stabilizing imaging payloads, and and orienting the solar panels toward the Sun [Paluszek, 2023]. Therefore, before any attempt to launch a spacecraft or even design or aquire the hardware sensors and actuators, an accurate

simulation of the ADCS in the global view of the mission is required to have a preliminary insight into what performance is expected.

Summary

This chapter introduces CubeSats as a revolutionary platform in space technology, highlighting their cost-effectiveness, modular design, and widespread adoption. It emphasizes the critical role of Attitude Determination and Control Systems (ADCS) in ensuring mission success, particularly for Earth observation tasks. The chapter sets the stage for the thesis by outlining the need for accurate ADCS simulations to predict performance before hardware implementation.

Chapter 2

Mission

2.1 Objectives

During this thesis, an attitude control system (ACS) will be developed for a hypothetical CubeSat in orbit, hereafter known as NPUSat, short for Northwestern Polytechnical University Satellite. The launch platform is the Tiangong Space Station. This means that the orbit of NPUSat is approximately the same as that of Tiangong. The mission of NPUSat will be to track ground locations during its orbit around Earth, for space-based Earth observation. For that, NPUSat is equipped with reaction wheels for detumbling after jettison and for attitude maneuvering.

A distinction should be made between orbit determination and control and attitude determination and control. This work deals with the latter of the two. Attitude refers to the orientation/rotation of the satellite in space. It is assumed that the parameters describing the orbit, explained in Subsection 5.1.1, are determined beforehand and no attempt will be made to e.g. change the altitude of the satellite, which is orbit control. This is a valid assumption because orbit parameters for all spacecraft are determined by the North American Aerospace Defense Command (NORAD) and are published in the Two Line Element (TLE) format, also described in Subsection 5.1.2. This means that the position of the satellites can be estimated directly, at a given time, without measurements from satellite on-board sensors. However, this requires frequent updates of the orbit parameters in order to maintain accuracy over time, see discussion in Subsection 5.1.3.

2.2 System Overview

In the following section, an overview of the ADCS will be provided. The ADCS consists of a controller, an actuator system, and a sensor feedback loop. The controller processes the attitude error, calculated from the reference input and the measured output. It then generates a control command for the reaction wheel actuator, which adjusts the satellite's attitude dynamics. External disturbances may affect the system, introducing deviations that the controller must compensate for. A sensor continuously monitors the attitude response and feeds it back to the controller for real-time corrections. A general block diagram of the control scheme is shown in Figure 2.1. Note that attitude determination will not be implemented in this project, and a simplified simulation of its mechanisms will be discussed in chapter 6.

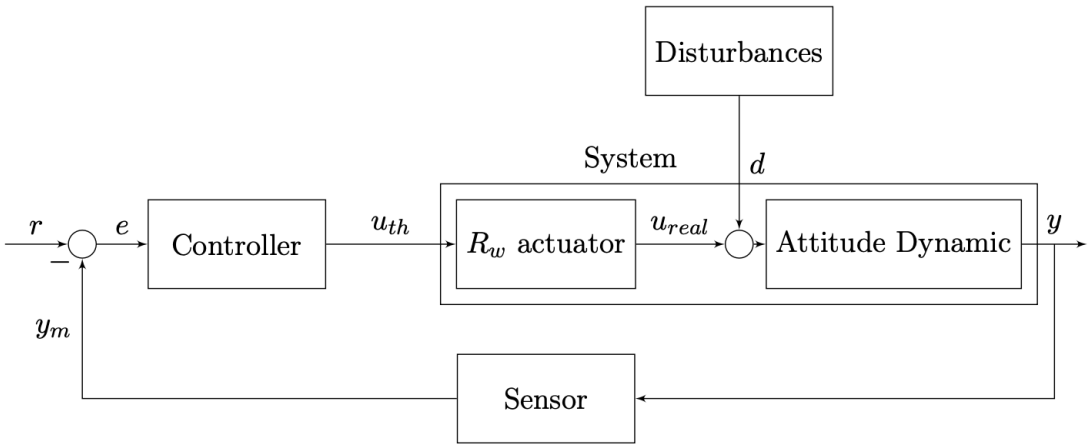


Figure 2.1: System block diagram of the ADCS.

2.3 ACS Operation Modes

The functionality of the ACS is broken down into a set of five modes, used in different phases of the spacecraft's mission. The transitions between these modes will, to some extent, be autonomous. A subset of the ACS mode transitions will be controllable manually from a ground station during a pass or by the use of a flight planner inside the spacecraft. In the following, the different modes are introduced.

2.3.1 Orbit Insertion

The Orbit Insertion mode is active immediately after jettison from Tiangong. During the first 45 minutes after jettison, the CubeSat is not allowed to generate or transmit any signal [CalPoly, 2022]. This means that the ACS will not be allowed to use magnetic actuation. The reaction wheels will not be spun up until the spacecraft has a tumble-rate below the threshold defined by the requirements in Chapter 4. The resulting attitude and possible tumbling rate are therefore provided by the launch vehicle, until after 45 minutes.

Functional Requirements:

Requirement	Description
Orbit Insertion	There are no requirements for this mode.

Table 2.1: Requirements for the Orbit Insertion.

2.3.2 Detumble

When the Orbit insertion phase is finished and the spacecraft powers up, the satellite will be tumbling with a rate given by Tiangong and the accumulated disturbances during the first phase of the spacecraft mission. This means that the spacecraft will need to perform some Detumbling. This maneuver will eventually slow the spacecraft's angular velocity down to a point where a mode transition to a more advanced attitude control mode is performed.

Functional Requirements:

Requirement	Description
Rate	Must be able to detumble from 10 [deg/s] to 1 [deg/s] in three orbits.

Table 2.2: Requirements for the Detumble mode.

2.3.3 Nadir

The Nadir mode is the mode used when the spacecraft is idle. Idle in the sense that the spacecraft does not have any targets to track, or is collecting data for the Earth's surface directly below it, in effect, pointing at the Earth's center, as shown in Figure 2.2.

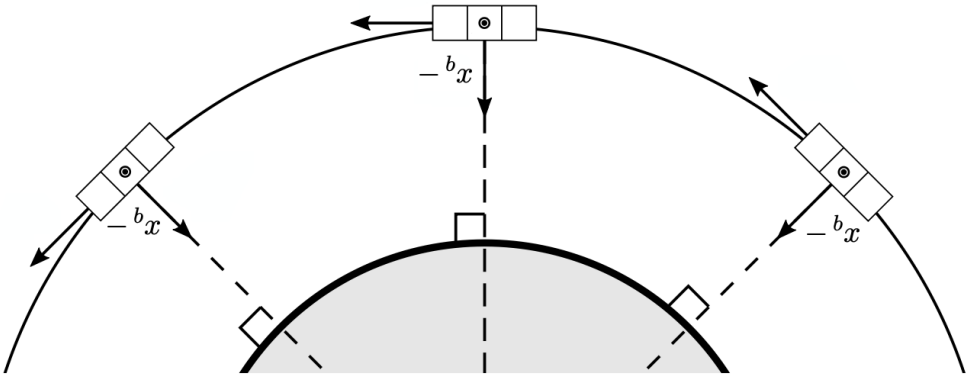


Figure 2.2: Shows the satellite orientation when in Nadir mode.

Because the orbit of NPUSat is circular, the angular velocity can be described as a function of the orbital period P which is defined by the orbital parameters [Wertz & Larson, 1999]. The angular velocity is then given as:

$$\omega_n = \frac{2\pi}{P} \tag{2.1}$$

The orbital period P in the lifetime of NPUSat is found to range from 92.56 min. at 400 km down to 90.52 min. at 300 km [Wertz & Larson, 1999]. It is calculated by:

$$P = \frac{2\pi}{60}\sqrt{\frac{(R_E + h)^3}{\mu}} \tag{2.2}$$

The largest angular velocity required for Nadir pointing is:

$$\omega_n = \max\left\{\frac{2\pi}{92.56 \cdot 60}, \frac{2\pi}{90.52 \cdot 60}\right\} = 0.0012 \quad \text{[rad/s]} \tag{2.3}$$

Functional Requirements:

Requirement	Description
Accuracy	Must be able to point $<$ 1 [deg] error
Slew rate	Must support attitude changes at angular velocities $>$ 0.0012 [rad/s]
Range	Must be able to attain Nadir attitude from all initial attitudes

Table 2.3: Requirements for the Nadir mode.

2.3.4 Target Tracking

The target tracking mode is used when the spacecraft is performing active pointing, that is, when a specific target on the Earth's surface needs to be tracked.

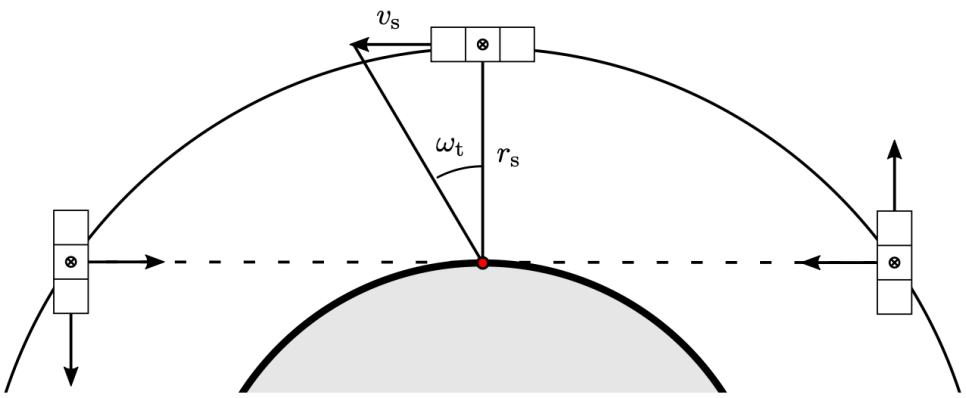


Figure 2.3: Shows the satellite orientation when in Target Tracking mode.

From Figure 2.3 it is seen that the angular rate of the spacecraft is a composition of the angular rate of pointing at Nadir and the angular rate necessary for tracking the target on the Earth surface. The lesser radius of the rotation sphere causes a higher angular velocity. Therefore the angular velocity necessary is calculated as:

$$\omega_n = \tan^{-1}\left(\frac{v_s}{r_s}\right) \tag{2.4}$$

where

ω_n	is the worst-case angular velocity	[[rad/s]]
v_s	is the orbital velocity of the satellite	[m/s]
r_s	is the smallest target vector length	[m]

The largest angular velocity seen from equation 2.4 is where the minimum length of the target vector rs is given. The orbit of NPUSat in its lifetime will experience a minimum distance to the Earth of 300 km at a speed of 7.726 km/s which results in

$$\omega_n = \tan^{-1}\left(\frac{7.726}{300}\right) = 0.0257\,[[rad/s]] \tag{2.5}$$

Functional Requirements:

Requirement	Description
Accuracy	Must be able to point $<$ 0.5 [deg] error during tracking
Slew rate	Must support attitude changes at angular velocities $>$ 0.00257 [rad/s]
Responsiveness	Must be able to track targets in under 1 minute

Table 2.4: Requirements for the Target Tracking mode.

2.3.5 Contingency

The contingency mode is a mode used when the spacecraft experiences a severe event. This event could be if an actuator partially breaks down or if the ADS degrades significantly, e.g., in eclipse. For this purpose, the Detumble mode is reused.

Functional Requirements:

Requirement	Description
Contingency Readiness	Must be able to re-enter Detumble mode from any state within 10 seconds
Reliability	Must be able to function with a set of 3 reaction wheels

Table 2.5: Requirements for the Contingency mode.

2.4 Operating the ACS Modes

These modes are modeled as a finite state machine where each state represents a mode. A general state transition diagram of the ACS states can be seen in Figure 2.4.

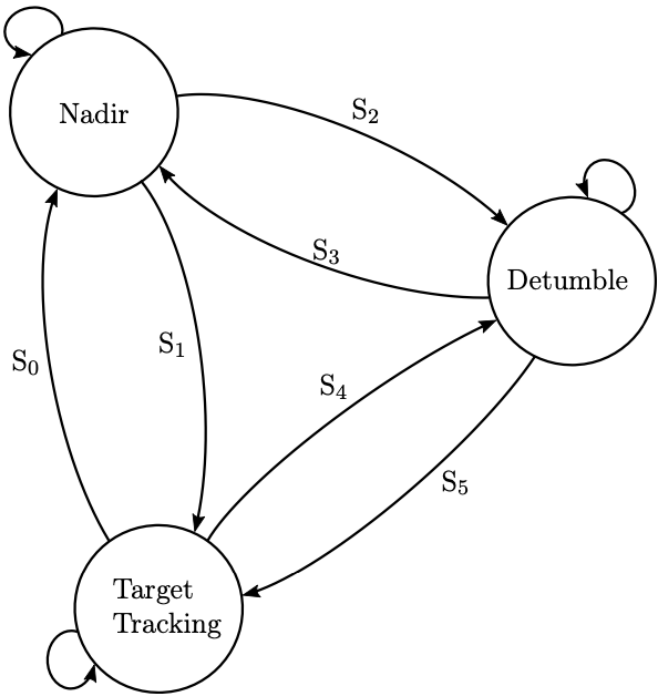


Figure 2.4: State transition diagram of the ACS.

When the spacecraft enters the LEOP it boots up in the Detumble mode. When the the spacecraft's tumble rate are below an allowed threshold, a state transition occurs, into Nadir

mode - from which the spacecraft waits for a command from the ground operator on where to go from here. The state transition conditions from Figure 2.4 are listed below:

S0 is the transition from the active target tracking mode to the Nadir mode. This transition will typically occur when a pass has ended or the object being tracked is not in line of sight anymore.

S1 is the transition from Nadir mode to the active target tracking mode. This transition occurs when the spacecraft has been idle and a target is within line of sight.

S2 is the transition between Nadir mode and Detumble mode. This transition can occur if the spacecraft experiences some event which triggers the contingency mode. This could be breakdown or partial breakdown of the actuators or if the ADS degrades.

S3 is the transition from the Detumble mode to the Nadir mode. This is the transition to occur automatically when the tumble rate of the spacecraft falls below the tumble threshold defined by the requirements. This is the default transition to happen from the Detumble mode, when the spacecraft has detumbled.

S4 is the transition from the Target tracking mode to the Detumble mode. This transition can occur if the spacecraft experiences some event which triggers the contingency mode. This could be breakdown or partial breakdown of the actuators or if the ADS degrades.

S5 is the transition from the Detumble mode to the Target tracking mode. This transition is not the default behaviour when the spacecraft has been detumbled but will only happen if the ground operator has expressively commanded it.

All of these state transitions can be triggered by either an on-board flight planner or a ground operator.

Summary

Chapter 2 defines the mission objectives of NPUSat, a hypothetical CubeSat designed for Earth observation. It details the satellite's operational modes—Orbit Insertion, Detumble, Nadir Pointing, Target Tracking, and Contingency—along with their respective requirements. The chapter also presents a system overview, including the control scheme and state transitions, ensuring compliance with mission constraints.

Chapter 3

Attitude Description

3.1 Reference Frames

Sources: [Markley & Crassidis, 2014]

To describe the attitude of a rigid body in three-dimensional space, one has to first define the reference frame used for this description. A reference frame is a representative term for a right-handed three-dimensional Cartesian coordinate system, described by three mutually perpendicular unit vectors. Several reference frames are of special interest for satellite attitude analysis, and when carefully placed, they simplify the kinematic and dynamic equations of the satellite. In this section, we will discuss the most important of these.

3.1.1 Earth-centered Inertial (ECI)

The specific ECI frame used throughout this thesis is the J2000 frame, defined with the Earth's mean equator and mean equinox at 12:00 terrestrial time on 1 January 2000. Accordingly, the origin of the ECI frame is fixed at the Earth's CoM. The ${}^{i}z$-axis is aligned with the Earth's rotational axis, the ${}^{i}x$-axis points in the direction of the mean vernal equinox, and the ${}^{i}y$-axis completes the right-handed Cartesian coordinate system such that the (${}^{i}x$, ${}^{i}y$)-plane spans the equatorial plane, as shown in Figure 3.1.

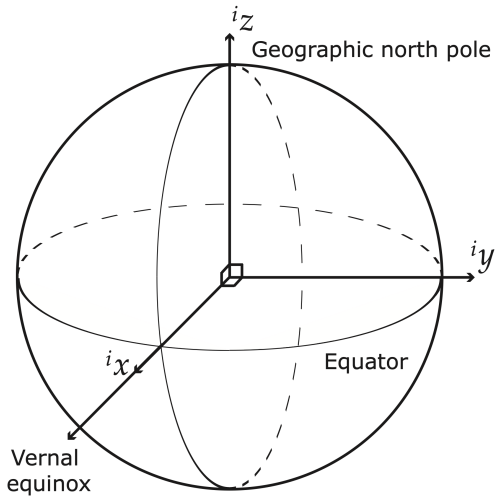


Figure 3.1: The ECI reference frame, where the ^{i}x-axis points towards the Vernal equinox.

3.1.2 Earth-centered Earth-fixed (ECEF)

The ECEF frame serves as a coordinate system that follows the rotation of the Earth. The origin of the ECEF frame is also fixed at the Earth's CoM, and the ^{e}z-axis likewise aligned with the Earth's rotational axis. The ^{e}x-axis points in the direction of the prime meridian perpendicular to the ^{e}z-axis, with the Greenwich Meridian being the internationally agreed-upon standard. The ^{e}y-axis completes the right-handed Cartesian coordinate system and the (^{e}x, ^{e}y)-plane also spans the equatorial plane, as shown in Figure 3.2.

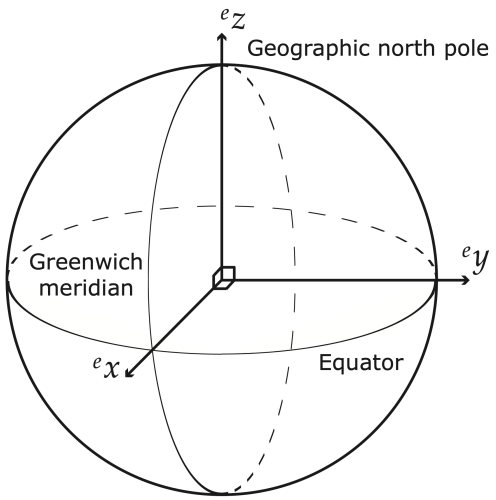


Figure 3.2: The ECEF reference frame, where the ^{e}x-axis is fixed to the Greenwich meridian.

3.1.3 Orbit Reference Frame (ORF)

The ORF maintains its orientation relative to the Earth and is centered in the satellite's CoM. The ${}^{o}z$-axis is always pointing in the direction of the Earth' s CoM (Nadir pointing), the ${}^{o}x$-axis points in the direction of the velocity vector of the spacecraft, and the ${}^{o}y$-axis is the cross product of the ${}^{o}x$- and ${}^{o}z$-axis, see Figure 3.3. The orientation of the satellite with respect to this system of coordinates is also known as roll (rotation about the ${}^{o}x$-axis), pitch (rotation about the ${}^{o}y$-axis), and yaw (the rotation about the ${}^{o}z$-axis).

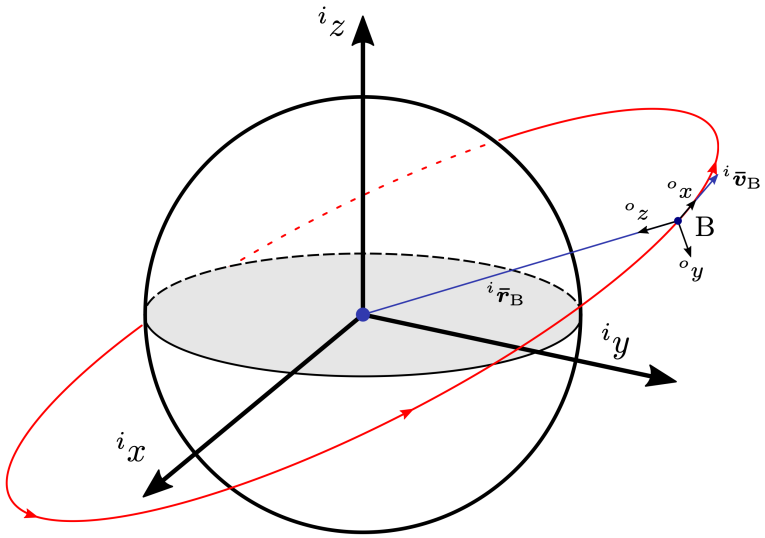


Figure 3.3: The ORF reference frame, where the ${}^{o}z$-axis is always pointing Nadir.

3.1.4 Target Reference Frame (TRF)

The purpose of the TRF is to track a point of interest on the surface of the Earth. The point is only visible from the spacecraft when it is over the local horizon plane, which is given as the tangent plane to the specific point. The origin of the TRF is defined at the spacecraft' s CoM. The ${}^{t}z$-axis of the TRF is the negative unit vector of the position vector denoted by $-{}^{\boldsymbol{i}}\vec{\boldsymbol{r}}_{\boldsymbol{\Delta}}$. The calculation of ${}^{\boldsymbol{i}}\vec{\boldsymbol{r}}_{\boldsymbol{\Delta}}$ is described in Section 6.6XXXXX. The ${}^{t}y$-axis is given as a unit vector perpendicular to the plane spanned by the velocity vector, parallel to ${}^{\boldsymbol{o}}\vec{\boldsymbol{x}}$ and ${}^{t}z$. The ${}^{t}x$-axis is then given as the cross product between ${}^{t}z$ and ${}^{t}y$ to complete the right-handed Cartesian coordinate system as shown in Figure 3.4.

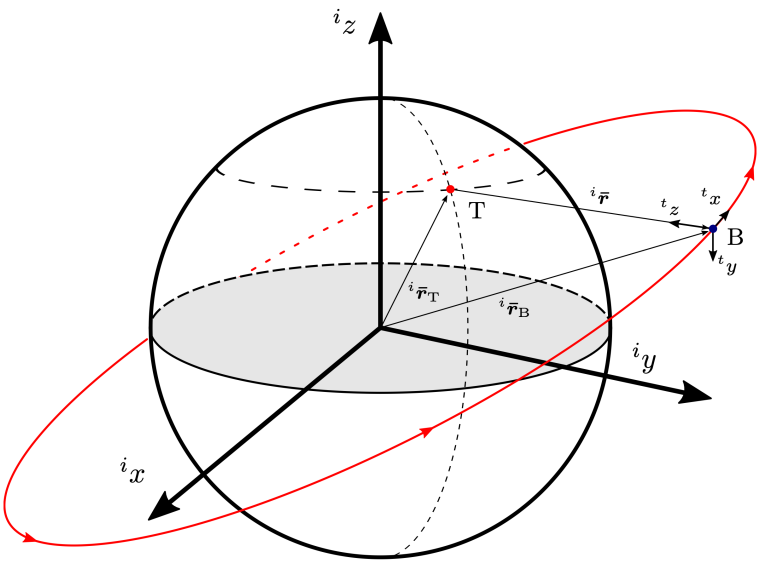


Figure 3.4: The TRF reference frame, where the ^{t}z-axis points to the target T and ^{t}x-axis points in the direction of flight.

3.1.5 Body Reference Frame (BRF)

The BRF is defined with its origin at the spacecraft's CoM. Its orientation is aligned according to the satellite's mission; for example, a payload containing a camera would have the ^{b}x-axis pointing in the direction of the camera. Accordingly, the ^{b}y-axis and ^{b}z-axis are placed perpendicular to the satellite sides. The attitude error of a spacecraft in orbit can be characterized by the angular difference between the TRF and the BRF, representing the rotation necessary to point the camera at a specific location on the Earth's surface. In this project, the BRF is defined as the coordinate system shown in Figure 3.5.

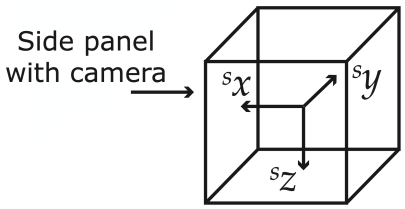


Figure 3.5: The BRF reference frame, where the ^{b}x-axis points in the direction of the camera.

3.1.6 Control Reference Frame (CRF)

The CRF is used to describe the satellite's mass distribution and has its origin in the spacecraft's CoM. All the external torques from the actuators and disturbances are expressed in this reference frame. The CRF axes are aligned with the principal axes of the spacecraft' s inertia matrix: The ${}^{c}z$-axis points in the direction of the major principal axis, corresponding to the largest moment of inertia (MoI); the ${}^{c}x$-axis aligns with the minor principal axis, where the MoI is smallest; and the ${}^{c}y$-axis is the perpendicular to the (${}^{c}x$, ${}^{c}z$)-plane, completing the right-handed Cartesian coordinate system.

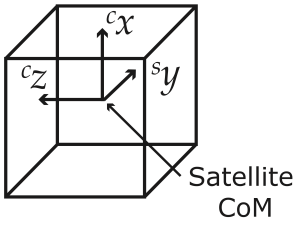


Figure 3.6: The CRF reference frame, centered in the CoM.

3.2 Rotations

Sources: [Markley & Crassidis, 2014]

3.2.1 Rotation Between ECI and ECEF

The rotation from ECI to ECEF is given by the rotation of the Earth. The ${}^{e}z$-axis rotates around the ${}^{i}z$-axis with a period time of a sidereal day denoted by t_d. This rotation is described by a direction cosine matrix (DCM) with a rotation around the z-axis.

$$ {}^{e}_{i}\underline{\mathbf{R}} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3.1} $$

where ${}^{e}_{i}\underline{\mathbf{R}}$ is the rotation matrix from ECI to ECEF and $\theta(t)$ is the angle between the ECI and ECEF and is given as:

$$ \theta(t) = \frac{2\pi \cdot t}{t_d} \tag{3.2} $$

Where t represents the time elapsed since the last alignment of the ECEF with the ECI frame. The direction cosine can be expressed as a quaternion, as shown below:

$$\begin{aligned}
{}^{e}_{i}q_4 &= \pm\frac{1}{2}\sqrt{1+R_{11}+R_{22}+R_{33}} = \sqrt{\frac{1+\cos(\theta)}{2}} \\
{}^{e}_{i}q_1 &= \frac{R_{23}-R_{32}}{4q_4} = 0 \\
{}^{e}_{i}q_2 &= \frac{R_{31}-R_{13}}{4q_4} = 0 \\
{}^{e}_{i}q_3 &= \frac{R_{12}-R_{21}}{4q_4} = \frac{-2\sin(\theta)}{4q_4}
\end{aligned} \tag{3.3}$$

It should be noted that when ${}^{e}_{i}q_4$ is small, calculations become inaccurate as described in Appendix A.2.

3.2.2 Rotation Between ECI and ORF

In order to find the rotation from ECI to ORF is derived from the position vector and velocity vector of the spacecraft given in the ECI frame as shown in Figure 3.3.

The ${}^{o}z$-axis of the ORF frame is defined to point in the direction of Nadir:

$${}^{o}\hat{\boldsymbol{z}} = -\frac{{}^{i}\vec{\boldsymbol{r}}_S}{\|{}^{i}\vec{\boldsymbol{r}}_S\|} \tag{3.4}$$

$${}^{o}\hat{\boldsymbol{x}} = \frac{{}^{i}\vec{\boldsymbol{v}}_B}{\|{}^{i}\vec{\boldsymbol{v}}_B\|} \tag{3.5}$$

Where ${}^{i}\vec{\boldsymbol{v}}_B$ is the velocity vector of the spacecraft given in ECI. Then ${}^{o}\hat{\boldsymbol{y}}$ is given as the cross product of ${}^{o}\hat{\boldsymbol{z}}$ and ${}^{o}\hat{\boldsymbol{x}}$.

$${}^{o}\hat{\boldsymbol{y}} = {}^{o}\hat{\boldsymbol{z}} \times {}^{o}\hat{\boldsymbol{x}} \tag{3.6}$$

The rotation matrix from ECI to ORF can then be written as:

$${}^{o}_{i}\underline{\mathbf{R}} = [{}^{o}\hat{\boldsymbol{x}}\ {}^{o}\hat{\boldsymbol{y}}\ {}^{o}\hat{\boldsymbol{z}}] \tag{3.7}$$

3.2.3 Rotation Between ECI and TRF

The rotation between ECI and TRF is derived from the position vector of the satellite and the target given in the ECI frame.

$${}^{t}\hat{\boldsymbol{z}} = -\frac{{}^{i}\vec{\boldsymbol{r}}_S - {}^{i}\vec{\boldsymbol{r}}_T}{\|{}^{i}\vec{\boldsymbol{r}}_S - {}^{i}\vec{\boldsymbol{r}}_T\|} \tag{3.8}$$

Where:

${}^{i}\vec{\boldsymbol{r}}_S$	is the satellite position vector	[m]
${}^{i}\vec{\boldsymbol{r}}_T$	is the target position vector	[m]

The ${}^{t}y$-axis is then derived from the velocity vector of the satellite given in the ORF. The ${}^{t}y$-axis of the TRF is defined as being perpendicular to the plane spanned by the velocity vector ${}^{i}\vec{\boldsymbol{v}}_B$ and the ${}^{t}\hat{\boldsymbol{z}}$.

$$ {}^{t}\hat{\boldsymbol{y}} = {}^{t}\hat{\boldsymbol{z}} \times \frac{{}^{i}\vec{\boldsymbol{v}}_B}{\|{}^{i}\vec{\boldsymbol{v}}_B\|} \tag{3.9} $$

where:

$${}^{i}\vec{\boldsymbol{v}}_B \quad \text{is the satellite velocity vector given in the ECI frame} \quad [\text{m/s}]$$

The ${}^{t}\hat{\boldsymbol{x}}$ of the TRF is then the normal to the $({}^{t}\hat{\boldsymbol{y}}, {}^{t}\hat{\boldsymbol{z}})$-plane forming a right-handed Cartesian coordinate system.

$$ {}^{t}\hat{\boldsymbol{x}} = {}^{t}\hat{\boldsymbol{y}} \times {}^{t}\hat{\boldsymbol{z}} \tag{3.10} $$

3.2.4 Rotation Between ECI and BRF

The attitude of the satellite is given by the rotation of the BRF relative to the ECI. The orientation of the spacecraft is found through an ADS.

3.2.5 Rotation Between BRF and CRF

It is possible for the CRF and the BRF to have zero rotation relative to each other, which would indicate that the satellite's mass is symmetrically distributed across its body. However, this is not always the case. In many instances, the satellite's center of mass (CoM) is offset from the geometric center, and the principal axes of the inertia matrix are rotated due to an uneven mass distribution.

The rotation caused by this uneven distribution of mass can be determined by performing an eigenvalue decomposition of the inertia matrix, which yields the corresponding eigenvectors and eigenvalues, as shown in the following equation for uneven mass distribution.

$$ \underline{\mathbf{J}} = \underline{\mathbf{R}}\,\underline{\mathbf{J}'}\,\underline{\mathbf{R}}^{-1} \tag{3.11} $$

Summary

Chapter 3 establishes the reference frames (ECI, ECEF, ORF, TRF, BRF, CRF) and mathematical representations (quaternions, rotation matrices) used to describe satellite attitude. It derives transformations between frames and explains the kinematic relationships governing rotations, providing the foundation for subsequent dynamic modeling and control design.

Chapter 4

Satellite Model

In this chapter, the kinematical and dynamical equations of the satellite are established, the external disturbances are identified, and the chosen actuators are modeled. Quaternion algebra, which will be used from this chapter onward, is thoroughly described in Appendix A.

4.1 Kinematical Equations of Motion

Sources: [Wertz, 1995] [Sidi, 1997] [Markley & Crassidis, 2014]

The kinematics of a satellite are defined as its motion over time. Since quaternions are used for attitude representation, the kinematics are then the time derivative of the attitude quaternion.

The attitude quaternion $\mathbf{q}(t + \Delta t)$ at the time $t + \Delta t$ can be split up into:

$$\mathbf{q}(t + \Delta t) = \mathbf{q}(t) \otimes \mathbf{q}(\Delta t) \tag{4.1}$$

Dividing the quaternion into a real and a pure quaternion part

$$\mathbf{q}(\Delta t) = \cos\left(\frac{\Delta\Phi}{2}\right) + \vec{\boldsymbol{u}} \sin\left(\frac{\Delta\Phi}{2}\right) \tag{4.2}$$

Where the sine and cosine parts can be replaced by the small angle approximation:

$$\cos\left(\frac{\Delta\Phi}{2}\right) \approx 1 \quad \text{and} \quad \sin\left(\frac{\Delta\Phi}{2}\right) \approx \frac{1}{2}\omega\Delta t \tag{4.3}$$

Equation (5.1) and equation (5.3) can then be combined into:

$$\mathbf{q}(t+\Delta t) \approx \left(\underline{\mathbf{I}} + \frac{1}{2}\omega\Delta t \begin{bmatrix} 0 & -u_3 & u_2 & u_1 \\ u_3 & 0 & -u_1 & u_2 \\ -u_2 & u_1 & 0 & u_3 \\ -u_1 & -u_2 & -u_3 & 0 \end{bmatrix} \right) \mathbf{q}(t) \tag{4.4}$$

This can be simplified to:

$$\mathbf{q}(t+\Delta t) \approx \frac{1}{2}\Delta t \cdot \underline{\mathbf{\Omega}}(\vec{\boldsymbol{\omega}})\mathbf{q}(t) + \mathbf{q}(t) \tag{4.5}$$

Where $\underline{\mathbf{\Omega}}(\vec{\boldsymbol{\omega}})$ is the four-dimensional skew-symmetric cross-product operator:

$$\underline{\mathbf{\Omega}}(\vec{\boldsymbol{\omega}}) = \begin{bmatrix} 0 & -\omega_3(t) & \omega_2(t) & \omega_1(t) \\ \omega_3(t) & 0 & -\omega_1(t) & \omega_2(t) \\ -\omega_2(t) & \omega_1(t) & 0 & \omega_3(t) \\ -\omega_1(t) & -\omega_2(t) & -\omega_3(t) & 0 \end{bmatrix} \tag{4.6}$$

Equation (5.5) is rearranged to:

$$\frac{\mathbf{q}(t+\Delta t) - \mathbf{q}(t)}{\Delta t} = \frac{1}{2}\underline{\mathbf{\Omega}}(\vec{\boldsymbol{\omega}})\mathbf{q}(t) \tag{4.7}$$

If a limit is imposed, the kinematic equation finally appears as:

$$\dot{\mathbf{q}}(t) = \lim_{\Delta t \to 0} \frac{\mathbf{q}(t+\Delta t) - \mathbf{q}(t)}{\Delta t} = \frac{1}{2}\underline{\mathbf{\Omega}}(\vec{\boldsymbol{\omega}})\mathbf{q}(t) \tag{4.8}$$

4.2 Dynamical Equations of Motion

Sources: [Wertz, 1995] [Sidi, 1997] [Markley & Crassidis, 2014]

The dynamics of a satellite describe its motion relative to the forces exerted. In the case of rotational motion, torques are of more importance, whether they stem from actuator torques or disturbance torques. The rotational dynamics are derived using Euler' s rigid body dynamics.

The angular momentum can be written as:

$${}^{i}\vec{\boldsymbol{L}}(t) = \underline{\mathbf{J}}_s \, {}^{i}\vec{\boldsymbol{\omega}}(t) \tag{4.9}$$

Where $\vec{\boldsymbol{\omega}}(t) = [\omega_1 \, \omega_2 \, \omega_3]^T$ and $\underline{\mathbf{J}}_s$ is the inertia matrix of the satellite.

According to Euler's second law, the applied torque can be derived as the time derivative of the angular momentum:

$$ {}^{i}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) = \frac{d}{dt}\, {}^{i}\vec{\boldsymbol{L}}(t) = {}^{i}\dot{\vec{\boldsymbol{L}}}(t) \tag{4.10} $$

Where the torque described in the control reference frame can be written as:

$$ \begin{aligned} {}^{c}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) &= \frac{d}{dt}\left({}^{c}_{i}\underline{\mathbf{R}}\, {}^{i}\vec{\boldsymbol{L}}(t)\right) \\ &= {}^{c}_{i}\dot{\underline{\mathbf{R}}}\, {}^{i}\vec{\boldsymbol{L}}(t) + {}^{c}_{i}\underline{\mathbf{R}}\, {}^{i}\dot{\vec{\boldsymbol{L}}}(t) \end{aligned} \tag{4.11} $$

Where the derivative of a rotation matrix can be written as:

$$ {}^{c}_{i}\dot{\underline{\mathbf{R}}} = -{}^{c}\vec{\boldsymbol{\omega}}(t) \times {}^{c}_{i}\underline{\mathbf{R}} \tag{4.12} $$

Inserted in equation (5.11) this gives:

$$ \begin{aligned} {}^{c}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) &= -{}^{c}\vec{\boldsymbol{\omega}}(t) \times {}^{c}_{i}\underline{\mathbf{R}}\, {}^{i}\vec{\boldsymbol{L}}(t) + {}^{c}_{i}\underline{\mathbf{R}}\, {}^{i}\dot{\vec{\boldsymbol{L}}}(t) \\ &= -{}^{c}\vec{\boldsymbol{\omega}}(t) \times {}^{c}\vec{\boldsymbol{L}}(t) + {}^{c}\dot{\vec{\boldsymbol{L}}}(t) \end{aligned} \tag{4.13} $$

Using equation (5.9) and isolating the angular acceleration, results in the satellite dynamic equation:

$$ \begin{aligned} {}^{c}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) &= -{}^{c}\vec{\boldsymbol{\omega}}(t) \times \underline{\mathbf{J}}_{s}\, {}^{c}\vec{\boldsymbol{\omega}}(t) + \underline{\mathbf{J}}_{s}\, {}^{c}\dot{\vec{\boldsymbol{\omega}}}(t) \\ \underline{\mathbf{J}}_{s}\, {}^{c}\dot{\vec{\boldsymbol{\omega}}}(t) &= -{}^{c}\vec{\boldsymbol{\omega}}(t) \times \underline{\mathbf{J}}_{s}\, {}^{c}\vec{\boldsymbol{\omega}}(t) + {}^{c}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) \\ {}^{c}\dot{\vec{\boldsymbol{\omega}}}(t) &= \underline{\mathbf{J}}_{s}^{-1}[-{}^{c}\vec{\boldsymbol{\omega}}(t) \times \underline{\mathbf{J}}_{s}\, {}^{c}\vec{\boldsymbol{\omega}}(t) + {}^{c}\vec{\boldsymbol{\tau}}_{\text{ext}}(t)] \end{aligned} \tag{4.14} $$

Equation (5.14) can be simplified to:

$$ {}^{c}\dot{\vec{\boldsymbol{\omega}}}(t) = \underline{\mathbf{J}}_{s}^{-1}[-\underline{\mathbf{S}}({}^{c}\vec{\boldsymbol{\omega}})\, \underline{\mathbf{J}}_{s}\, {}^{c}\vec{\boldsymbol{\omega}}(t) + {}^{c}\vec{\boldsymbol{\tau}}_{\text{ext}}(t)] \tag{4.15} $$

where ${}^{c}\vec{\boldsymbol{\tau}}_{\text{ext}}$ is the sum of all external torques; from both the actuators and the disturbances. $\underline{\mathbf{S}}({}^{c}\vec{\boldsymbol{\omega}})$ is the three-dimensional skew-symmetric crossproduct operator, defined as:

$$ \underline{\mathbf{S}}(\vec{\boldsymbol{\omega}}) = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \tag{4.16} $$

4.3 Actuator Models

To realize the operational modes of the NPUSat, and to counter the different types of disturbances when in orbit, the ACS is equipped with a set of reaction wheels. The models of the reaction wheels will be described throughout this section.

4.3.1 Reaction wheels

Sources: [Sidi, 1997], [Franklin, 2010], [Xia, 2012]

A three-wheeled configuration, where each reaction wheel is placed parallel to a principal axis of the body, simplifies the control problem by having direct translation of moment between reaction wheel momentum and the spacecraft's principal axes. There is, however, a risk of losing control of one axis if one of the wheels ceases to function, leading to complete mission failure. Consequently, a redundant configuration with four wheels is employed.

Dynamic Model

The momentum of the reaction wheels is generated by brushless DC (BLDC) motors. Unlike traditional DC motors, BLDC motors energize their windings based on the rotor's position and are controlled by a three-phase signal. For each phase, the winding generates electromagnetic torque and back-EMF, similar to a DC motor. This similarity allows the same models used for DC motors to be applied when modeling BLDC motors.

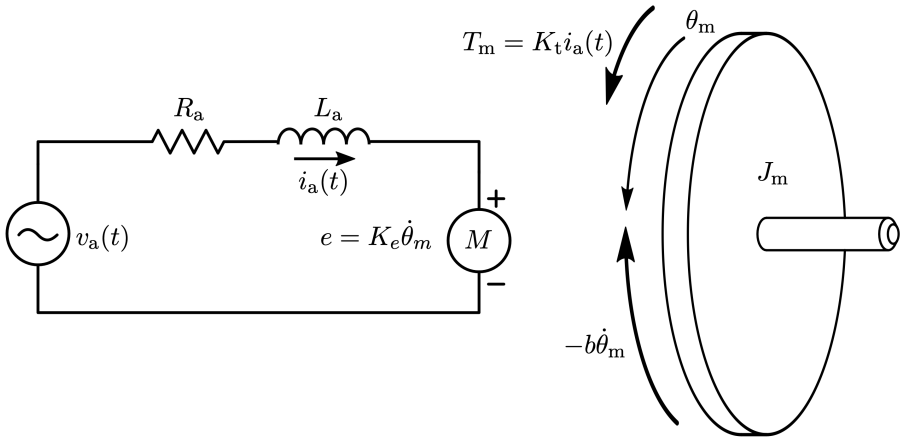


Figure 4.1: Shows the electric circuit of the armature to the left, and the free-body diagram of the rotor to the right.

The free-body diagram in Figure 4.1 shows the two torques affecting the rotor: T_m and the friction $-b\dot{\theta}_m$ acting in its opposite direction. The rotational equivalent of Newton's law of motion states that the sum of all external moments about the CoM of a rotating body is given as:

$$\sum \tau_{ext} = J\dot{\omega} \tag{4.17}$$

Where J is the one-dimensional inertia of the rotating body in the rotation axis and $\dot{\omega}$ is the

angular acceleration. The equation of angular motion for the mechanical part of the motor is given in the following equation:

$$J_m\dot{\omega}_m = K_t i_a - b\omega_m \tag{4.18}$$

Where:

K_t	is the motor torque constant	[Nm/A]
i_a	is the armature current	[A]
b	is the motor friction coefficient	[Nm·s/rad]

The electrical part of the motor as shown in Figure 4.1 is given as an RL circuit modeled as:

$$L_a\frac{d}{dt}i_a(t) + R_a\, i_a(t) = v_a(t) - K_e\,\omega_m \tag{4.19}$$

Where:

L_a	is the armature self inductance	[H]
R_a	is the armature electrical resistance	[Ω]
v_a	is the armature voltage	[V]
K_e	is the motor friction coefficient	[V·s/rad]

For a BLDC, the current running through the windings is controlled by three phases, with each phase separated by 120 degrees, resulting in the current being effectively only in two of the windings at any given time. This means that the motor coefficients should be measured phase-to-phase.

The dynamics of the electrical circuit induction relative to the dynamics of the mechanical motion is often said to have negligible effect and is therefore neglected. The combination of equation of motion from equation 5.18 and electrical dynamics from equation 5.19 is then given as equation 5.20:

$$J_m\,\dot{\omega}_m = -\left(\frac{K_t\,K_e}{R_a} + b\right)\omega_m \;+\; \frac{K_t}{R_a}\,v_a(t) \tag{4.20}$$

As seen in equation 4.20, $J_m\,\dot{\omega}_m = \tau_m$, which is the sum of all torques on the motor shaft. Then it is known from Newton' s third law of motion, that for every action there is an equal and opposite reaction. In the case of a rotating body, the action exerted on the motor shaft is the motor torque τ_m, and an equally large reaction $-\tau_m$ acts on the mount that holds the motor

(e.g., the satellite structure).

The reaction torque $-\tau_m$ is thereby directly translated to the satellite body if the motor is oriented along a principal axis of the spacecraft. To suppress disturbance torques and reorientate the satellite, the motors must be able to produce a sufficiently large τ_m. One way to achieve this is by increasing the inertia on the motor shaft. As seen from equation 5.20, the torque is directly dependent on the motor inertia J_m and the angular acceleration $\dot{\omega}_m$.

$$\tau_{rw} = -(J_{\mathrm{w}} + J_m)\dot{\omega}_m \tag{4.21}$$

where τ_{rw} is the reaction torque acting on the satellite body and J_{w} is the inertia of the reaction wheel. Since $J_{\mathrm{w}} \gg J_m$, the shaft inertia J_m is often neglected. The dimensioning of the reaction wheels also depends on the number of actuators used and their configuration.

Reaction Wheels Configuration

As described before, the use of four reaction wheels increases the reliability of the ACS by providing redundancy if one motor fails. In such a configuration, the wheels must be oriented so that the torque generated by each has a component on at least two of the three principal axes of the satellite.

The NPUSat attitude control is equipped with four reaction wheels in a tetrahedron configuration. This configuration offers redundancy, zero angular momentum transition capability, and can produce up to twice the maximum torque of a single wheel on any axis. It is chosen over the pyramid configuration for its relatively lower power consumption, either without or with a wheel failure [Kök, 2012].

All four wheels in the tetrahedron configuration are orientated such that when commanded to produce the same torque (e.g., to manage secular disturbances or for momentum dumping maneuvers against magnetorquers), they would produce no net torque on the spacecraft body, only changing their collective stored angular momentum. The distribution of the individual wheel torques τ_i (for $i \in [1, 4]$) in the three-dimensional control space is given by the distribution matrix $\underline{\mathbf{P}}_{\mathrm{w}}$, as shown in equation 4.22.

$$\begin{bmatrix} \tau_{cx} \\ \tau_{cy} \\ \tau_{cz} \end{bmatrix} = \underline{\mathbf{P}}_{\mathrm{w}} \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \end{bmatrix} \tag{4.22}$$

where $\underline{\mathbf{P}}_w$ is the tetrahedron configuration's distribution matrix, defined to have a row sum of zero -which means all torques generated by the reaction wheels will cancel out each other when running at the same biased angular velocity- and is given by:

$$\underline{\mathbf{P}}_{\mathrm{w}} = \begin{bmatrix} \cos\beta & -\cos\beta\cos\alpha & -\cos\beta\cos\alpha & 0 \\ 0 & \cos\beta\cos\frac{\alpha}{2} & -\cos\beta\cos\frac{\alpha}{2} & 0 \\ -\sin\beta & -\sin\beta & -\sin\beta & 1 \end{bmatrix} \tag{4.23}$$

with $\alpha = 60$ [deg] and $\beta = \arcsin(1/3) \approx 19.47$ [deg]. In this case, the X-Y projections of wheels 1, 2, and 3 are each separated by 120 degrees azimuthally. This provides a balanced distribution for the base of the tetrahedron.

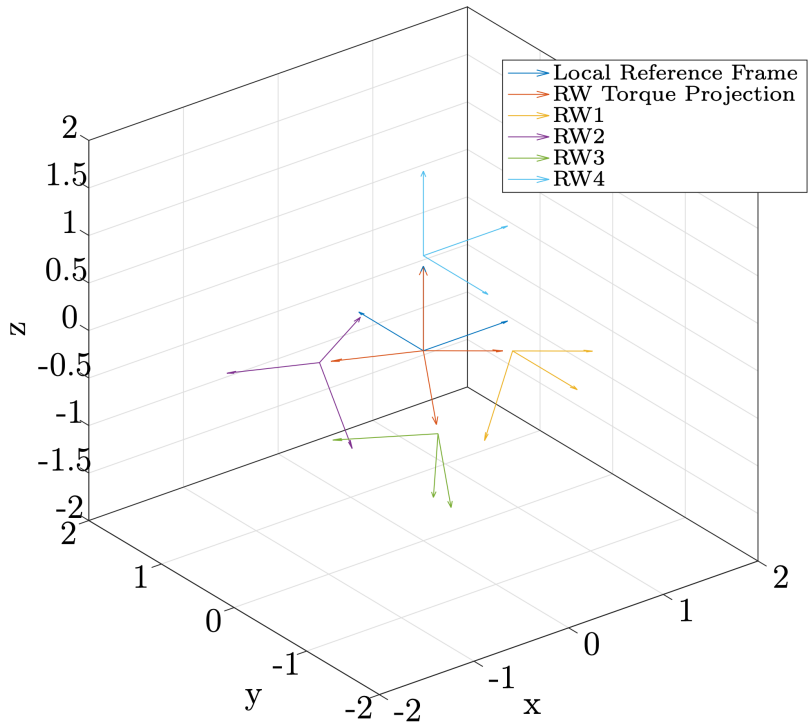


Figure 4.2: Tetrahedral reaction wheels configuration.

The $\vec{\boldsymbol{\tau}}_c$ vector is calculated from the control laws. Since there are four reaction wheels, the system is said to be over-actuated, meaning that there exists infinite solutions to the torque component τ_i. This is reflected in the distribution matrix $\underline{\mathbf{P}}_w$ which is an under-determined system.

Control Allocation

In order to find a solution to an over-actuated system given as:

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \end{bmatrix} = \underline{\mathbf{P}}_{\mathrm{w}}^{+} \begin{bmatrix} \tau_{cx} \\ \tau_{cy} \\ \tau_{cz} \end{bmatrix} \tag{4.24}$$

where $\underline{\mathbf{P}}_{\mathrm{w}}^{+}$ is the right Moore–Penrose pseudo-inverse of an under-determined system. The right pseudo-inverse of an under-determined system can be found as:

$$\underline{\mathbf{A}}^{+} = \underline{\mathbf{A}}^{T} \left(\underline{\mathbf{A}}\, \underline{\mathbf{A}}^{T} \right)^{-1} \tag{4.25}$$

where $\underline{\mathbf{A}}^{+}$ is the minimum norm solution to the over-actuated system, which means that the control torque will be evenly distributed among all four reaction wheels. Then the linear-map from the control torque $\vec{\boldsymbol{\tau}}_c$ given in the CRF to a desired reaction wheel torque is given as:

$${}^{w}\vec{\boldsymbol{\tau}} = \underline{\mathbf{P}}_{\mathrm{w}}^{+}\, {}^{c}\vec{\boldsymbol{\tau}}_c \tag{4.26}$$

Consequently, whenever a tree-dimensional control torque has been computed, there exists a linear combination that maps it to the four-dimensional reaction wheels torque, by the use of the right Moore-Penrose pseudo-inverse matrix.

4.3.2 Dynamic Model of the Configuration

The dynamic equations of reaction wheels mounted in the satellite frame are described by Euler's second law, which states that a change in angular momentum ${}^{i}\dot{\vec{\boldsymbol{L}}}(t)$ in an inertial reference frame is equal to the sum of all external torques ${}^{i}\vec{\boldsymbol{\tau}}_{\text{ext}}(t)$ acting on a rigid body, expressed as:

$${}^{i}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) = \frac{d}{dt}\, {}^{i}\vec{\boldsymbol{L}}(t) = {}^{i}\dot{\vec{\boldsymbol{L}}}(t). \tag{4.27}$$

Whereas the change in angular momentum of a single reaction wheel is given by the rotation of the angular momentum between the ECI frame and one Reaction Wheel Reference Frame (RWRF), via multiplication by the rotation matrix ${}^{w}_{i}\underline{\mathbf{R}}$. Symbolically,

$$
\begin{aligned}
{}^{w}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) &= \frac{d}{dt}\Big({}^{w}_{i}\underline{\mathbf{R}}\, {}^{i}\vec{\boldsymbol{L}}(t)\Big) \\
&= {}^{w}_{i}\dot{\underline{\mathbf{R}}}\, {}^{i}\vec{\boldsymbol{L}}(t) + {}^{w}_{i}\underline{\mathbf{R}}\, {}^{i}\dot{\vec{\boldsymbol{L}}}(t)
\end{aligned}
\tag{4.28}
$$

The derivative of the rotation matrix is expressed as the product of the skew-symmetric cross-product operator $\underline{\mathbf{S}}(\cdot)$ defined in equation 5.16 and the rotation matrix from ECI to RWRF:

$$
{}^{w}_{i}\dot{\underline{\mathbf{R}}} = -\underline{\mathbf{S}}({}^{w}\vec{\boldsymbol{\omega}}(t))\, {}^{w}_{i}\underline{\mathbf{R}} \tag{4.29}
$$

Substituting equation 5.29 into equation 5.28 yields the resulting expression for ${}^{w}\vec{\boldsymbol{\tau}}_{\text{ext}}(t)$ in terms of the wheel' s angular momentum and the time derivatives of the rotation matrix:

$$
\begin{aligned}
{}^{w}\vec{\boldsymbol{\tau}}_{\text{ext}}(t) &= -\underline{\mathbf{S}}({}^{w}\vec{\boldsymbol{\omega}}(t))\, {}^{w}_{i}\underline{\mathbf{R}}\, {}^{i}\vec{\boldsymbol{L}}(t) + {}^{w}_{i}\underline{\mathbf{R}}\, {}^{i}\dot{\vec{\boldsymbol{L}}}(t) \\
&= -\underline{\mathbf{S}}({}^{w}\vec{\boldsymbol{\omega}}(t))\, {}^{w}\vec{\boldsymbol{L}}(t) + {}^{w}\dot{\vec{\boldsymbol{L}}}(t)
\end{aligned}
\tag{4.30}
$$

Further than this, it can be seen that the change of angular momentum of a reaction wheel mounted on the satellite body is equal to the sum of the Coriolis force and the external forces:

$$
{}^{w}\dot{\vec{\boldsymbol{L}}}(t) = -\underline{\mathbf{S}}({}^{w}\vec{\boldsymbol{\omega}}(t))\, {}^{w}\vec{\boldsymbol{L}}(t) + {}^{w}\,\vec{\boldsymbol{\tau}}_{\text{ext}}(t) \tag{4.31}
$$

where ${}^{w}\vec{\boldsymbol{\tau}}_{\text{ext}}(t)$ is the torque generated by the BLDC motor and ${}^{w}\vec{\boldsymbol{\omega}}$ is the sum of the rotational velocity of the satellite in the RWRF and the reaction wheel given as:

$$
{}^{w}\vec{\boldsymbol{\omega}}(t) = {}^{w}\vec{\boldsymbol{\omega}}_{s}(t) + {}^{w}\vec{\boldsymbol{\omega}}_{\text{w}}(t) \tag{4.32}
$$

where ${}^{w}\vec{\boldsymbol{\omega}}_{s}(t) = [{}^{w}\vec{\boldsymbol{\omega}}_{sx}\, {}^{w}\vec{\boldsymbol{\omega}}_{sy}\, {}^{w}\vec{\boldsymbol{\omega}}_{sz}]^{T}$ and ${}^{w}\vec{\boldsymbol{\omega}}_{\text{w}}(t) = [0\,0\, {}^{w}\vec{\boldsymbol{\omega}}_{\text{wz}}]^{T}$.

The change in angular momentum ${}^{w}\dot{\vec{\boldsymbol{L}}}(t)$ is now described for one wheel in the RWRF. The torque from the reaction wheel is described in RWRF and is translated to the CRF where the MoI of the satellite is defined. Therefore, the change in angular momentum in the CRF caused by the reaction wheel is described as:

$$
{}^{c}\dot{\vec{\boldsymbol{L}}}_{\text{w}}(t) = -{}^{c}_{w}\underline{\mathbf{R}}\, {}^{w}\dot{\vec{\boldsymbol{L}}}_{\text{w}}(t) \tag{4.33}
$$

The sign change of the torque exerted on the CRF is explained by Newton' s third law of motion, that states, for every action acting on a rigid body, there exists an equally large opposite reaction, which also applies for rotational bodies. This implies that the change in angular momentum on the reaction wheel has an opposite effect on the satellite CRF.

The rotational dynamics described in equation 5.31 is only accompanying one reaction wheel. The total change in angular momentum is then given as the sum of all moments generated by the reaction wheels. It should be noted that for every reaction wheel, a unique rotation matrix exists and is denoted by $i \in [1; 4]$:

$$^{c}\dot{\vec{\boldsymbol{L}}}_{\text{th}}(t) = -\sum_{i=1}^{4} {}^{c}\dot{\vec{\boldsymbol{L}}}_{\text{w},i}(t) \tag{4.34}$$

The resulting change in angular momentum $^{c}\dot{\vec{\boldsymbol{L}}}_{\text{th}}(t)$ is then added to the external torques in the dynamic model of the satellite given in Section 5.2.

4.4 Disturbances Models

Space is a harsh environment featuring multiple disturbances. The major disturbances taken into account in low-earth orbits are aerodynamic drag, residual magnetic torque, gravity-gradient torque, and radiation pressure. In the following sections these four disturbances will be analyzed.

4.4.1 Aerodynamic Drag

Sources: [NASA, 1971]

In the upper atmosphere at orbital heights for a LEO, gas particles collide with the surface of the satellite. This creates a force on the spacecraft opposite to the direction of travel.

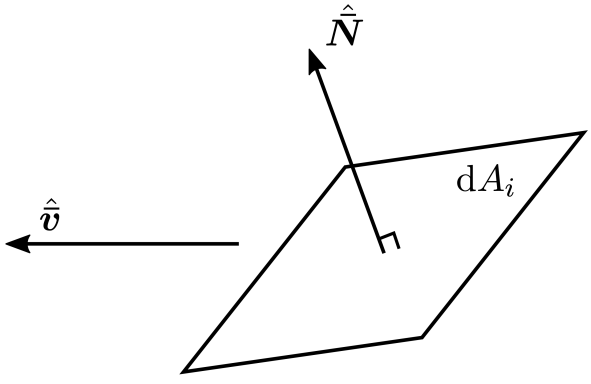


Figure 4.3: The vectors involved in calculating the aerodynamic drag.

The force created is dependent on the size of the incident area, the atmospheric density, the travel velocity and the drag coefficient of the incidence surface. The i-th aerodynamic force

$d\vec{\boldsymbol{F}}_{D,i}$ on the i-th infinitesimal small surface element dA_i, can be calculated as:

$$d\vec{\boldsymbol{F}}_{D,i} = -\frac{1}{2} C_D \, \rho \, v^2 \left[\hat{\boldsymbol{N}}^T \hat{\boldsymbol{v}} \right] \hat{\boldsymbol{v}} \, dA_i \quad [\text{N}] \tag{4.35}$$

where:

C_D	is the drag coefficient	$[-]$
ρ	is the atmospheric density	[kg.m^{-3}]
v	is the orbital velocity	[m.s^{-1}]
A	is the incidence area	[m^2]
$\hat{\boldsymbol{N}}$	is a unit vector normal to the incident area	$[-]$
$\hat{\boldsymbol{v}}$	is a unit vector in the velocity direction	$[-]$

The aerodynamic drag coefficient C_D is usually between 1 and 2. A good estimate is $C_D = 2$ in cases where the drag coefficient is unknown. The expected atmospheric density ρ at an altitude of 400 [km] is $1.54{\times}10^{-12}$ [kg.m$^-$3], as computed using the NRLMSIS 2.1 empirical atmospheric model via NASA's Community Coordinated Modeling Center (CCMC) Instant Run tool [NASA CCMC, 2022].

It is important to notice that the atmospheric density is not constant, but varies as a function of e.g. solar activity and whether the air is in Sunlight or not [Hughes, 2004]. However, the air density will be modeled as constant in this thesis.

Assuming that the satellite can be seen as a plane, with the area A_s and an outward normal unit vector $\hat{\boldsymbol{N}}$ parallel to the translational velocity unit vector $\hat{\boldsymbol{v}}$, then the aerodynamic force $\vec{\boldsymbol{F}}_D$ acting on the satellite is given as:

$$\vec{\boldsymbol{F}}_D = -\frac{1}{2} C_D \, \rho \, v^2 \, \hat{\boldsymbol{v}} \, dA_{sat} \quad [\text{N}] \tag{4.36}$$

The aerodynamic torque $\vec{\boldsymbol{\tau}}_D$ affecting the satellite is then given as:

$$\vec{\boldsymbol{\tau}}_D = \vec{\boldsymbol{r}}_{CoM \rightarrow \, CoP} \times \vec{\boldsymbol{F}}_D \quad [\text{Nm}] \tag{4.37}$$

where $\vec{\boldsymbol{r}}_{CoM \rightarrow \, CoP}$ is a vector from the CoM to the Center of Pressure CoP of the satellite.

4.4.2 Residual Magnetic Torque

Sources: [NASA, 1969 March]

A satellite in orbit, travels in the geomagnetic field of the Earth. This leads to three main magnetic disturbance torques. These are the effects of the interaction between the satellite and geomagnetic field.

- Eddy currents.
- Hysteresis.
- Spacecraft magnetic moments.

When a ferromagnetic material is exposed to magnetism, then an electric field is induced in the material. This electric field leads to small currents running in the ferromagnetic material, called Eddy Currents.

Hysteresis is the phenomenon of ferromagnetic materials on a molecular level gets magnetised by another magnet. When exposed to a magnetic field, the molecules in the material will align with the field and thereby rendering its own magnetic field.

Spacecraft Magnetic Moments; of the three disturbances mentioned here, this is usually the largest one. Spacecraft generated magnetic moments are caused by, for example, the wiring inside the satellite, the Printed Circuit Board (PCB) layout and everywhere else, where a current loop is present.

Eddy Currents and Hysteresis are negligible, compared to the Spacecraft Magnetic Moment and in the following only the latter will be considered.

The spacecraft magnetic moment $\vec{\boldsymbol{\tau}}_{mag}$ can be calculated as:

$$\vec{\boldsymbol{\tau}}_{mag} = \vec{\boldsymbol{m}} \times \vec{\boldsymbol{B}} \quad \text{[Nm]} \tag{4.38}$$

where:

$\vec{\boldsymbol{m}}$	is the residual dipole moment of the satellite	[A.m^2]
$\vec{\boldsymbol{B}}$	is the geomagnetic field vector	[T]

The magnetic residual torque $\vec{\boldsymbol{\tau}}_{mag}$ is inherently difficult to calculate because of the uncertainty in residual dipoles direction and size, and if compensated for it may be negligible. In [Wertz, 1995], the Firesat example is suggested to have a magnetic moment $\vec{\boldsymbol{m}}$ of 1 [A.m^{-2}] for a 110 [W] satellite. A single unit CubeSat is expected to have an average power of 1 [W] which is approximately 100 times less than the Firesat. This would give a qualified guesstimate of 0.01 [A.m^{-2}] for NPUSat's residual dipole moment $\vec{\boldsymbol{m}}$.

4.4.3 Gravity-Gradient Torque

Sources: [NASA, 1969 May]

A non-symmetrical satellite orbiting the Earth, or any satellite in a non-uniform gravitational field, will experience a gravitational torque. This torque, often called the gravity-gradient torque, arises because the gravitational force exerted by the Earth varies across the satellite's structure. Since gravity is inversely proportional to the square of the distance, different parts of the satellite at slightly different altitudes or orientations will experience slightly different gravitational pulls. If the Earth's gravitational field were perfectly uniform, or if the satellite were perfectly spherically symmetrical in its mass distribution, no net gravitational torque would act on it.

Assuming that the satellite is only influenced by the Earth' s gravitational field, the satellite consists of a single body, and that both the Earth and the satellite can be modeled as two point masses, it is possible to calculate the gravitational force $d\vec{\boldsymbol{F}}_{G,i}$ exerted by the Earth on an infinitesimal mass element of the satellite using newton' s law of gravitation:

$$d\vec{\boldsymbol{F}}_{G,i} = -\mu_E \frac{\vec{\boldsymbol{R}}_{E,i} dm_i}{\|\vec{\boldsymbol{R}}_{E,i}\|^2} \quad \text{[N]} \tag{4.39}$$

where:

μ_E	is the Earth's standard gravitational parameter	[m^3.s^{-2}]
$\vec{\boldsymbol{R}}_{E,i}$	is the vector from the Earth's CoM to the i-th mass element's CoM	[m]
dm_i	is the mass of the i-th element	[kg]

The torque contribution from the infinitesimal mass element about the satellite's CoM is then found as:

$$d\vec{\boldsymbol{\tau}}_{G,i} = \vec{\boldsymbol{r}}_{GC\rightarrow\, CoM,i} \times d\vec{\boldsymbol{F}}_{G,i} \quad \text{[Nm]} \tag{4.40}$$

where:

$\vec{\boldsymbol{r}}_{GC\rightarrow\ CoM,i}$ is a vector from the satellite's geometric center to the element's CoM.

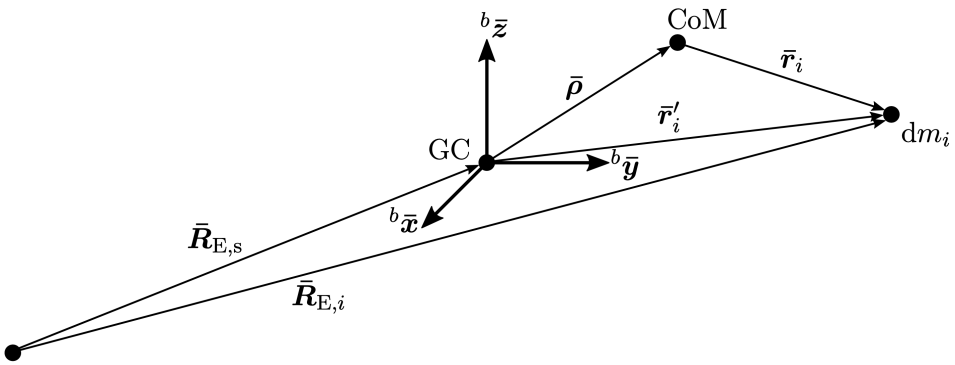


Figure 4.4: The vectors involved in calculating the gravity gradient torque, where GC is the geometric center of the satellite.

The resulting torque affecting the entire satellite can then be written as:

$$\vec{\boldsymbol{\tau}}_G = \int \vec{\boldsymbol{r}}_{GC\rightarrow\ CoM,i} \times d\vec{\boldsymbol{F}}_{G,i} = \int \vec{\boldsymbol{r}}_{GC\rightarrow\ CoM,i} \times -\mu_E \frac{\vec{\boldsymbol{R}}_{E,i} dm_i}{\|\vec{\boldsymbol{R}}_{E,i}\|^2} \quad \text{[Nm]} \tag{4.41}$$

If the satellite's CoM and geometric center are assumed identical, the equation can be simplified to:

$$\vec{\boldsymbol{\tau}}_G = \frac{3\mu_E}{\|\vec{\boldsymbol{R}}_{E,s}\|^3} \left[\hat{\boldsymbol{R}}_{E,s} \times \left(\underline{\mathbf{J}}_s \cdot \hat{\boldsymbol{R}}_{E,s} \right) \right] \quad \text{[Nm]} \tag{4.42}$$

where:

$\underline{\mathbf{J}}_s$	is the moment of inertia tensor of the satellite	[kg.m^3]
$\vec{\boldsymbol{R}}_{E,s}$	is the vector from the Earth's CoM to the satellite CoM	[m]

From equation 4.4.3 it can be seen that it produces a vector which the gravity gradient torque is about, and a scaling of that vector which depends on the distance to the Earth's center. If a spacecraft is in an elliptic orbit then the gravity gradient would vary over the course of the orbit; caused by the varying distance to the center of the Earth. The gravity gradient torque also depends on the mass distribution of the spacecraft as the cross product is between $\hat{\boldsymbol{R}}_{E,s}$ and $\underline{\mathbf{J}}_s \cdot \hat{\boldsymbol{R}}_{E,s}$ where the latter is a perturbation caused by the spacecraft's MoI tensor. If the spacecraft had a perfect evenly distributed MoI, then there would be no torque disturbance from the gravity gradient.

4.4.4 Radiation Pressure

Sources: [NASA, 1969 October]

A satellite in orbit is exposed to electromagnetic radiation pressure. The sources of this radiation pressure are:

- Direct Solar Radiation.
- Earth and Moon Albedo.
- Radiation from the Earth and its atmosphere.

Out of these, direct solar photon radiation is the most significant. The forces caused by the other sources (reflected Sunlight, and Earth's own thermal emissions) are usually at least an order of magnitude smaller. The intensity of solar corpuscular radiation (the solar wind, streams of charged particles, primarily protons and electrons, continuously emitted from the Sun's corona) is usually negligible.

A sufficient method of modeling the force affecting the satellite due to solar radiation, is to assume that the incident radiation is either absorbed, reflected specularly or reflected diffusely.

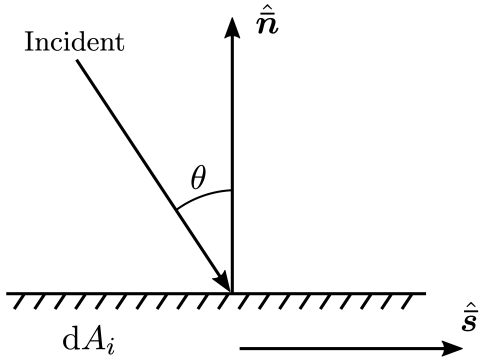


Figure 4.5: Absorption.

Radiation force for a completely absorbing surface:

$$d\vec{\boldsymbol{F}}_{a,i} = -\frac{I_{Sun}}{c} C_a \cos(\theta_{Sun})\, \hat{\boldsymbol{R}}_{S\rightarrow\, Sun}\, dA_i \quad \text{[N]} \quad (4.43)$$

where C_a = absorption coefficient (the fraction of the incident radiation that is absorbed). For this case,

$$C_a = 1,\ C_{rs} = 0,\ C_{rd} = 0.$$

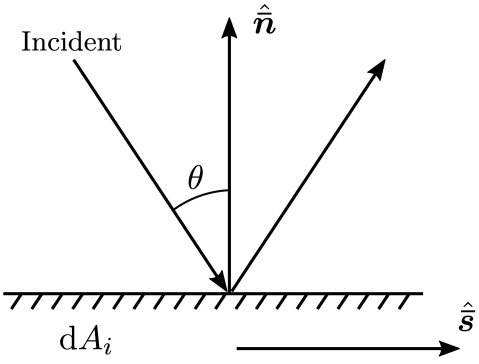


Figure 4.6: Reflection.

Radiation force for a specularly reflecting surface:

$$d\vec{\boldsymbol{F}}_{rs,i} = -2\frac{I_{Sun}}{c} C_{rs} \cos^2(\theta_{Sun}) \hat{\boldsymbol{n}}\, dA_i \quad [\mathrm{N}] \qquad (4.44)$$

where C_rs = specularly reflection coefficient (the fraction of the incident radiation that is specularly reflected). For this case, $C_a = 0$, $C_{rs} = 1$, $C_{rd} = 0$.

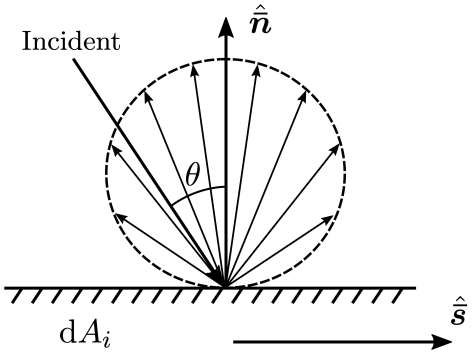


Figure 4.7: Diffusion.

Radiation force for a diffusely reflecting surface:

$$d\vec{\boldsymbol{F}}_{rd,i} = \frac{I_{sol}}{c} C_{rd} \left(-\frac{2}{3} \cos(\theta_{Sun}) \hat{\boldsymbol{n}} - \cos(\theta_{Sun}) \hat{\boldsymbol{R}}_{S\rightarrow\, Sun} \right) dA_i \,[\mathrm{N}] \qquad (4.45)$$

where C_rd = diffusely reflection coefficient (the fraction of the incident radiation that is diffusely reflected). For this case, $C_a = 0$, $C_{rs} = 0$, $C_{rd} = 1$.

where:

I_{sol}	is the total solar irradiance (TSI)	[W.m^{-2}]
c	is the speed of light in vacuum	[m.s^{-1}]
$\hat{\boldsymbol{n}}$	is a unit vector normal to the incident area	[m]
$\hat{\boldsymbol{R}}_{S\rightarrow\, Sun}$	is a unit vector in the direction from the satellite' s CoM to the Sun' s CoM.	[m]
θ_{Sun}	is the angel between $\hat{\boldsymbol{R}}_{S\rightarrow\, Sun}$ and $\hat{\boldsymbol{n}}$ where $(0 \leq \theta_{Sun} \leq \frac{\pi}{2})$	[rad]

A TSI of 1366 [W.m^{-2}] has been chosen. This value is the mean value of eight measurements obtained from 8 different satellites [Butler, 2008].

It is also possible to model the incident radiation as a combination of the previous mentioned cases, the radiation force for the ith surface area $\vec{\boldsymbol{F}}_{R,i}$ can be calculated as:

$$\vec{\boldsymbol{F}}_{R,i} = \int d\vec{\boldsymbol{F}}_{a,i} + d\vec{\boldsymbol{F}}_{rs,i} + d\vec{\boldsymbol{F}}_{rd,i} \quad [\mathrm{N}] \qquad (4.46)$$

It is common to simplify the irradiated surface of the satellite using simple geometrical elements, e.g. a plane. Assuming that the satellite can be described as a plane with the area

A_{sat}, then the solar radiation force exerted on the satellite $\vec{\boldsymbol{F}}_R$ is given as

$$\vec{\boldsymbol{F}}_R = -\frac{I_{sol}}{c}\left[(1-C_{rs})\hat{\boldsymbol{R}}_{S\rightarrow\, Sun} + 2\left(C_{rs}\cos(\theta_{Sun}) + \frac{1}{3}C_{rd}\right)\hat{\boldsymbol{n}}\right]\cos(\theta_{Sun})\,A_{sat} \quad [\mathrm{N}] \tag{4.47}$$

with:

$$0 \leq C_a + C_{rs} + C_{rd} \leq 1 \tag{4.48}$$

Knowing that the cross-sectional area of A_{sat} is perpendicular to $\hat{\boldsymbol{R}}_{S\rightarrow\, Sun}$ i.e. $\hat{\boldsymbol{n}} = \hat{\boldsymbol{R}}_{S\rightarrow\, Sun}$, it is possible to simplify the previous equation to:

$$\begin{aligned}
\vec{\boldsymbol{F}}_R &= -\frac{I_{sol}}{c}\left((1-C_{\text{rs}})\hat{\boldsymbol{R}}_{S\rightarrow\text{Sun}} + 2\left(C_{\text{rs}} + \frac{1}{3}C_{\text{rd}}\right)\hat{\boldsymbol{n}}\right)A_{\text{sat}} \\
&= -\frac{I_{sol}}{c}\left(\hat{\boldsymbol{R}}_{S\rightarrow\text{Sun}} - C_{\text{rs}}\left(\hat{\boldsymbol{R}}_{S\rightarrow\text{Sun}} - 2\hat{\boldsymbol{n}}\right) + \frac{2}{3}C_{\text{rd}}\hat{\boldsymbol{n}}\right)A_{\text{sat}} \\
&= -\frac{I_{sol}}{c}\left(1 + C_{\text{rs}} + \frac{2}{3}C_{\text{rd}}\right)A_{\text{sat}}\,\hat{\boldsymbol{R}}_{S\rightarrow\text{Sun}} \\
&= -\frac{I_{sol}}{c}\,C_{\text{rk}}\,A_{\text{sat}}\,\hat{\boldsymbol{R}}_{S\rightarrow\text{Sun}} \quad [\mathrm{N}]
\end{aligned} \tag{4.49}$$

The constant C_{rk} is used to specify the outer material of the satellite. A value of C_{rk} < 1 specifies a translucent material, C_{rk} = 1 specifies a perfect absorbent material (black body), and C_{rk} >1 specifies a reflecting material, where C_{rk} is in the range $0 \leq C_{rk} \leq 2$.

The radiation torque exerted on the satellite $\vec{\boldsymbol{\tau}}_R$ is then given as:

$$\vec{\boldsymbol{\tau}}_R = \vec{\boldsymbol{r}}_{CoM\rightarrow\, CoP} \times \vec{\boldsymbol{F}}_R \quad [\mathrm{Nm}] \tag{4.50}$$

Summary

Chapter 4 formulates the satellite's kinematic and dynamic equations of motion using quaternion algebra and Euler's rigid-body dynamics. It models reaction wheel actuators, including BLDC motor dynamics and tetrahedral configuration, and control allocation strategies for over-actuated systems. The chapter concludes with a derivation of disturbance torques (aerodynamic drag, magnetic, gravity-gradient, solar radiation).

Chapter 5

Environment Model

This chapter starts by giving a description of the different Keplerian orbit elements used to identify the orbit of a satellite around the Earth. Correspondingly, the Simplified General Perturbation (SGP4) orbit model will be described. Section 6.2 presents the identified disturbance models.

5.1 Orbit Model

5.1.1 Keplerian Orbits

Sources: [Wertz, 1995] [Sidi, 1997] [Markley & Crassidis, 2014]

According to Kepler's laws of planetary motion, an orbit around a planet traces the shape of an ellipse, with the planet being orbited lying at one of the ellipse's two foci.

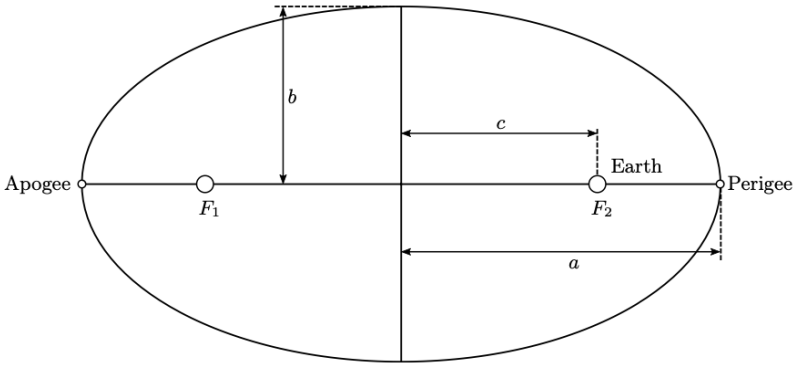


Figure 5.1: Geometric interpretation of a Keplerian orbit. $F1$ and $F2$ are the two foci of the ellipse, a and b are respectively semi-minor and semi-major axis.

The semi-major axis a and the eccentricity e define the size and shape of the ellipse respectively. The relationship between them is:

$$e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \sqrt{1 - \frac{b^2}{a^2}} \tag{5.1}$$

It is clear from Equation 5.1 that for a circle (c = 0), the eccentricity is 0. For an elliptic orbit, the eccentricity is $0 < e < 1$. Note that eccentricity can have values higher than 1, for what are called hyperbolic orbits, but which are not relevant to our work.

The elliptic orbit shown in Figure 5.1, can be expanded and used to describe an orbit in three-dimensional space, as illustrated in Figure 5.2, through the following orbital parameters:

- i, the inclination.
- Ω, the Right Ascension of the Ascending Node (RAAN).
- ω, the argument of perigee.

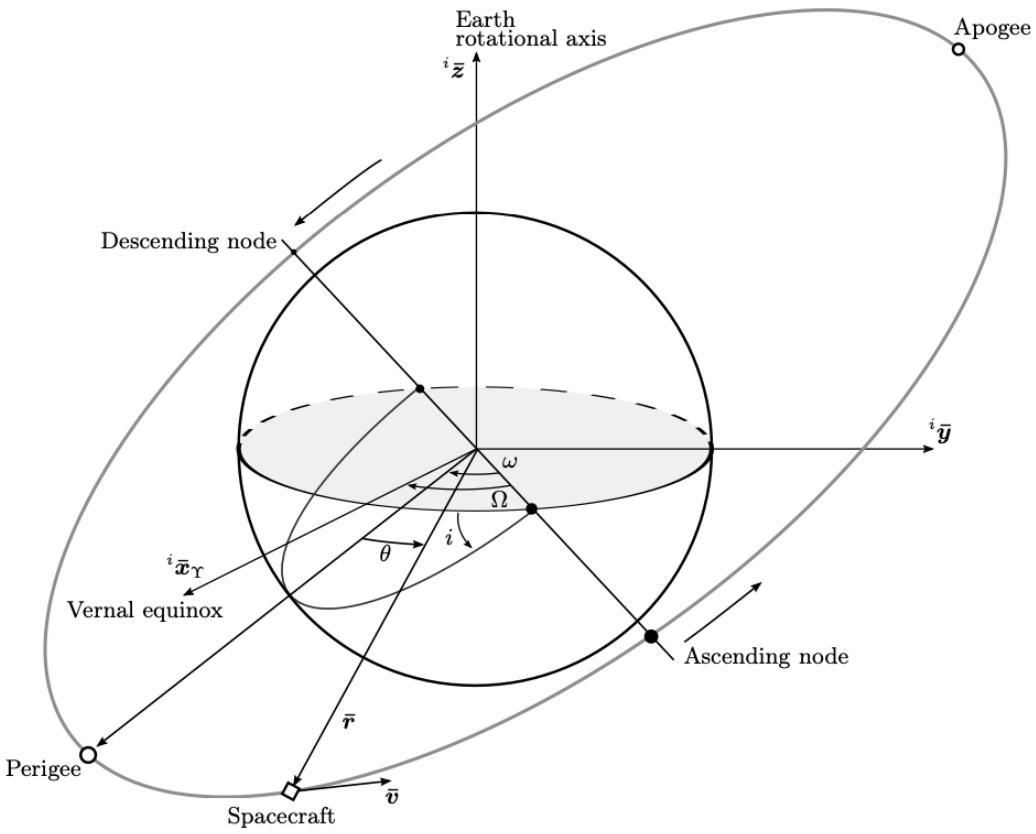


Figure 5.2: Keplerian orbit around the Earth specified using the classical orbital elements.

The **inclination** $\boldsymbol{i}$ is the angle between the orbital plane and the equatorial plane. Orbits with $i < 90$ [deg] are called **prograde** orbits, while orbits with $i > 90$ [deg] are called **retrograde** orbits. Orbits with $i = 90$ [deg] and $i = 0$ [deg] are called **polar** and **equatorial** orbits respectively.

The intersecting line between the orbital plane and the equatorial plane is called the **line of nodes**, and the point where the spacecraft intersects with this line traveling in the north direction, is called the **ascending node**; the opposite node where the spacecraft is traveling south is called the descending node. The **RAAN Ω** is the angle between the vernal equinox and the ascending node, measured **counterclockwise** in the equatorial plane.

The **apogee** and **perigee** denote respectively the points in orbit with the highest and lowest orbital altitudes. The **argument of perigee** $\boldsymbol{\omega}$, is the angle between the ascending node and the perigee of an orbit, measured counterclockwise in the orbital plane.

5.1.2 Two Line Elements

Sources: [Amsat, 2002], [CelesTrak, 2025], [Hoots, 1980]

An option to describe a satellite's motion after its launch is to use the Two Line Elements (TLE). This is based on the fact that every object that orbits the Earth and is bigger than a tennis ball, is tracked and monitored by radar by NORAD (North American Aerospace Defense Command, a bi-national organization between the United States of America and Canada). A TLE consists of two 69-character strings of orbital data, formatted as illustrated in Figure 5.3.

Line number	Satellite number	Classification	International designator: Year	Launch no.	Piece	Epoch time: Year	Julian date in current year [JD]	First derivative of mean motion [rev/day^2]	Second derivative of mean motion [rev/day^3]	BSTAR drag term [Earth_radii^{-1}]	Ephemeris type	Element no.	Checksum
1	2 3 4 5 6 7	8	9 10 11 12	13 14 15	16 17	18 19 20	21 22 23 24 25 26 27 28 29 30 31 32	33 34 35 36 37 38 39 40 41 42 43	44 45 46 47 48 49 50 51 52	53 54 55 56 57 58 59	60 61	62 63	64 65 66 67 68	69
1	d d d d d	c	d d	d d d	d d d	d d	d d d d . d d d d d d d d	s . d d d d d d d d	s . d d d d d s	s . d d d d d s	e	d	d d d d	d

Line number	Satellite number	Inclination i_o [Deg]	RAAN Ω_a [Deg]	Eccentricity e_e [-]	Argument of perigee ω_p [Deg]	Mean anomaly M_a [Deg]	Mean motion n_m [rev/day]	Epoch revision	Checksum
1	2 3 4 5 6 7	8 9 10 11 . 12 13 14 15 16	17 18 19 20 . 21 22 23 24 25	26 . 27 28 29 30 31 32 33	34 35 36 37 . 38 39 40 41 42	43 44 45 46 . 47 48 49 50 51	52 53 54 . 55 56 57 58 59 60 61	62 63 64 65 66 67	68
2	d d d d d	d d d . d d d d	d d d . d d d d	. d d d d d d d	d d d . d d d d	d d d . d d d d	d d . d d d d d d d d	d d d d d	d

Figure 5.3: TLE format where d is decimal number, c is character, and s is symbol

Classification can either be U (unclassified) or C (classified).

Epoch t_e specifies the time when the orbital parameters given in the TLE were obtained. It is in the Julian Date (JD) format. The JD is measured in the number of days since noon Universal Time (UT) -a modern continuation of the Greenwich Mean Time (GMT), which is the same everywhere on the Earth- on January 1, 4713 BC.

Mean Anomaly θ represents the angle describing the satellite's position in its orbital path. The angle is measured from perigee, where it is zero, and referenced to a circular orbit with radius equal to the orbits semi-major axis.

Mean Motion n represents the number of revolutions a satellite completes around Earth per day [rev/day]. For an unperturbed Keplerian orbit, it is defined as:

$$n = \sqrt{\frac{\mu}{a^3}} \tag{5.2}$$

where μ is Earth's standard gravitational parameter ($3.986004418 \times 10^{14}$ [$m^3.s^{-2}$]) and a is the semi-major axis of the orbit.

1.st Derivative of Mean Motion $\dot{n}$ given in the TLE as half the rate of change of the mean motion [rev/day^2]. This term primarily accounts for the effect of atmospheric drag. As drag reduces the orbital altitude, the satellite's velocity increases, resulting in a positive $\dot{n}$ value.

2.nd Derivative of Mean Motion $\ddot{n}$ represents the second time derivative of mean motion [rev/day^3]. While included in the TLE format for completeness, this term is typically set to zero in practice, as its effect is negligible for most orbital predictions.

Drag Term or B^*, is a dimensionless parameter that quantifies the effect of atmospheric drag on a satellite's orbit. It' s derived from the satellite' s shape, mass, and the atmospheric density at its altitude.

Epoch revision indicates the total number of orbits the satellite has completed around Earth since launch. Each revolution is counted from one ascending node crossing to the next. The count is maintained by tracking authorities and may not always start from zero.

5.1.3 Simplified General Perturbation Model

Sources: [Miura, 2009]

While numerous orbit propagation models exist, this work focuses primarily on the Simplified General Perturbations (SGP) family of models, particularly SGP, SGP4 and SGP8 for near-Earth orbits, along with their deep-space counterparts SDP4 and SDP8. These analytical models efficiently compute position and velocity vectors of Earth-orbiting objects by solving a set of *simplified* differential equations, avoiding the computational expense of numerical integration methods that require stepwise propagation from epoch.

A commonly used orbit propagator for LEO satellites is the SGP4. It not only considers the force of gravity from the Earth, but also orbital decay from drag, and effects caused by the Earth's oblateness and gravity from the Sun and the Moon. It is used thus in the simulations of NPUSat.

5.2 Sun Model

The Sun model is used to simulate the movement of the Sun in the ECI frame for attitude determination. Although it is in fact the Earth which orbits around the Sun, in the ECI frame the Sun appears to change position while the Earth is centered at the origin. The Sun model includes the same parameters and equations as the orbit model, described in Section 5.1. However, many of the parameters needed to calculate the position of the Sun in the ECI frame are well known and do not change significantly over time. This simplifies the model and reduces the necessary calculations.

The Epoch Mean Anomaly $M_{SunEpoch}$ of the Sun is given for the 1st of January 2000 at noon UTC, just like the orbit model uses a Julian Date referring from 2000 as input JD_{2000}. Since the Mean Motion of the Sun n_{Sun} is well known, this can be used to describe the current Mean Anomaly M_{Sun}, as shown in equation 5.3:

$$M_{Sun} = M_{SunEpoch} + n_{Sun} JD_{2000} = 357.528 + 0.9856003\, JD_{2000} \tag{5.3}$$

The position of the Sun in the ECI frame in the orbital plane (the ecliptic longitude) is given by equation 5.4. To calculate the ecliptic longitude of the Sun, most of the parameters characterizing the orbital motion of the Sun are already included in the equation. Thus, the only inputs needed are the Julian Date JD_{2000} and the current Mean Anomaly M_{Sun}.

$$V_{Sun} = 280.461 + 0.9856474\, JD_{2000} + 1.915 \sin(M_{Sun}) + 0.020 \sin(2M_{Sun}) \tag{5.4}$$

To transfer the position in the orbital plane into the ECI frame requires the tilt of the Earth' s rotational axis, which also represents the Z-axis of the ECI frame. This parameter is known as the obliquity of the ecliptic plane and is given in equation 5.5:

$$\epsilon = 23.4393 - 0.0000004\, JD_{2000} \tag{5.5}$$

The position of the Sun is then determined as a unit vector in the ECI frame by transforming the ecliptic longitude of the Sun V_{Sun} into Cartesian coordinates, including the obliquity ϵ of the ecliptic plane, as shown in equation 5.6:

$${}^{i}\hat{\boldsymbol{r}}_{Sun} = \begin{bmatrix} \cos(V_{Sun}) \\ \cos(\epsilon)\sin(V_{Sun}) \\ \sin(\epsilon)\sin(V_{Sun}) \end{bmatrix} \tag{5.6}$$

5.2.1 Eclipse Calculation

Sources: [Srivastava, 2013]

Knowing when a spacecraft is eclipsed is critical for thermal control, solar array pointing, and power management. Additionally, to accurately model the forces acting on the satellite due to solar radiation, it is important to know exactly when the satellite enters or exits a shadow region, since the amount of solar radiation incident on the spacecraft drops dramatically.

Eclipse prediction falls into two categories: spherical Earth conical shadow model and spherical Earth cylindrical shadow model. The first approximates the Earth' s shadow as an infinite cylinder aligned with the Sun' s direction, this method yields accurate results for low-earth orbits. The second accounts for the Earth's finite radius so that the shadow is conical, referred to as umbra, with the addition of a region that is partially illuminated called penumbra.

Since NPUSat orbits at low altitudes, the former method is opted for.

Earth Cylindrical Shadow Model

Assuming parallel Sunlight (due to the Sun being effectively infinitely distant from Earth), Earth casts a cylindrical shadow of radius equal to the Earth's mean radius r_e. Eclipse calculation then simplifies to a geometrically intuitive problem solvable via vector projections and dot products.

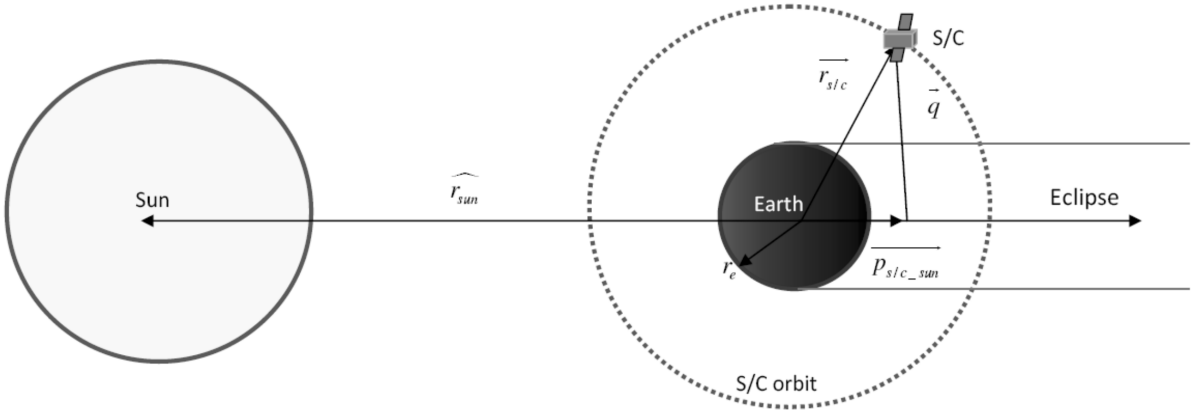


Figure 5.4: Earth' s shadow cylindrical geometry.

Given the spacecraft position vector ${}^i\vec{\boldsymbol{r}}_S$ and Sun unit vector ${}^i\hat{\boldsymbol{r}}_{Sun}$, the projection onto the Sun line is:

$$\vec{\boldsymbol{p}}_{S\rightarrow\, Sun} = ({}^i\vec{\boldsymbol{r}}_S \cdot {}^i\hat{\boldsymbol{r}}_{Sun}){}^i\hat{\boldsymbol{r}}_{Sun} \tag{5.7}$$

The orthogonal component from the shadow axis is:

$$\vec{\boldsymbol{q}} = {}^i\vec{\boldsymbol{r}}_S - \vec{\boldsymbol{p}}_{S\rightarrow\, Sun} \tag{5.8}$$

with magnitude:

$$\|\vec{\boldsymbol{q}}\| = \sqrt{\|{}^i\vec{\boldsymbol{r}}_S\|^2 - ({}^i\vec{\boldsymbol{r}}_S \cdot {}^i\hat{\boldsymbol{r}}_{Sun})^2} \tag{5.9}$$

Then, the illumination state follows these conditions:

- If ${}^i\vec{\boldsymbol{r}}_S \cdot {}^i\hat{\boldsymbol{r}}_{Sun} > 0$, then the spacecraft lies in the Sun-facing hemisphere and is in the Sunlit state.
- If ${}^i\vec{\boldsymbol{r}}_S \cdot {}^i\hat{\boldsymbol{r}}_{Sun} < 0$ and $||\vec{\boldsymbol{q}}|| < r_E$, where r_E is the Earth's mean radius, then the spacecraft lies behind the Earth and within the umbral region, i.e., it is in eclipse.

- Otherwise, the spacecraft is outside the Earth's shadow cone and hence in Sunlit state.

This method allows efficient eclipse prediction using basic linear algebra, making it well-suited for real-time onboard processing and robust against numerical instabilities associated with trigonometric functions.

5.3 Magnetic field model

An accurate representation of the Earth's magnetic field is crucial for simulating disturbance torques caused by a satellite's residual magnetic moment. The precision of this magnetic field model directly influences the reliability of attitude disturbance predictions.

5.3.1 The Earth's Magnetic Field

Sources: [Kane, 1983]

The Earth's magnetic field originates from multiple sources, with the dominant contribution —over 90%—coming from dynamo action in the Earth's liquid outer core. This component, known as the Main Field, evolves slowly over time and can be approximated by a dipole field. In this approximation, the Earth behaves like a giant bar magnet with magnetic poles located near the geographic poles, and magnetic field lines extending into space.

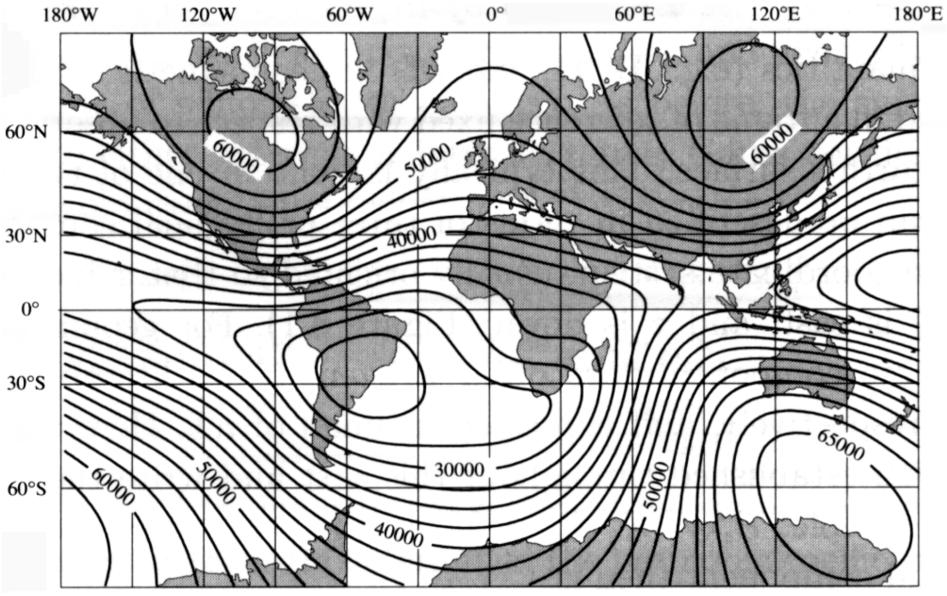


Figure 5.5: Magnetic field intensity at the Earth's surface in nanoTeslas.

In addition to the Main Field, smaller-scale and more rapidly varying magnetic components arise from electric currents in the ionosphere, magnetosphere, and Earth's crust. These external and crustal fields contribute to short-term temporal and spatial variations in the total geomagnetic field.

To model the Earth's magnetic environment, standardized geomagnetic field models are used. Two widely adopted models are the International Geomagnetic Reference Field (IGRF), developed and maintained by the International Association of Geomagnetism and Aeronomy (IAGA), and the World Magnetic Model (WMM), jointly released by the U.S. National Geophysical Data Center (NGDC) and the British Geological Survey (BGS).

In this simulation environment, the IGRF model has been selected due to its higher resolution and scientific focus. Its implementation and configuration will be detailed in the subsequent subsection.

5.3.2 The International Geomagnetic Reference Field

Sources: [Korhonen, 2000] [Roithmayr, 2004]

The International Geomagnetic Reference Field (IGRF) was developed and maintained by the International Association of Geomagnetism and Aeronomy (IAGA). It is constructed using a global set of geomagnetic observations collected from ground-based observatories, surveys, and satellite missions. The model expresses the Main Field as the negative gradient of a scalar potential, which is formulated using a truncated spherical harmonic expansion. This expansion is characterized by a set of coefficients, updated approximately every five years to reflect the slow temporal evolution of the field. The current version, IGRF-14, is valid for the epoch 2025–2030 and includes both the main field coefficients and their secular variation (i.e., the annual rate of change). The IGRF assumes linear variation within each five-year interval.

The magnetic field vector ${}^{e}\vec{\boldsymbol{B}}$ can be expressed as a sum of spherical harmonics ${}^{e}\vec{\boldsymbol{B}}_{n,m}$, as given in equation 5.10.

$$ {}^{e}\vec{\boldsymbol{B}} = \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} {}^{e}\vec{\boldsymbol{B}}_{n,m} \tag{5.10} $$

The contribution of spherical harmonics are of degree n and order m, and is given by equation 5.11.

$$
{}^{e}\vec{\boldsymbol{B}}_{n,m} = \frac{K_{n,m}a^{n+2}}{R^{n+m+1}} \Bigg\{ \frac{g_{n,m}C_m + h_{n,m}S_m}{R} \left[(sin(\lambda)A_{n,m+1} + (n+m+1)A_{n,m})\, {}^{e}\vec{\boldsymbol{r}} - A_{n,m+1}\, {}^{e}\hat{\boldsymbol{z}} \right] \\
- mA_{n,m} \left[(g_{n,m}C_{m-1} + h_{n,m}S_{m-1})\, {}^{e}\hat{\boldsymbol{x}} + (h_{n,m}C_{m-1} - g_{n,m}S_{m-1})\, {}^{e}\hat{\boldsymbol{y}} \right] \Bigg\} \tag{5.11}
$$

where:

a	Mean radius of the Earth	[km]
R	Magnitude of the position vector from Earth's center	[km]
${}^{e}\vec{\boldsymbol{r}}$	Unit vector in the direction of R	[-]
n	Degree of contributing spherical harmonics	[-]
m	Order of contributing spherical harmonics	[-]
$K_{n,m}$	Schmidt-to-Legendre coefficient	[-]
$A_{n,m}$	Derived Legendre polynomial of degree n and order m	[-]
λ	Geographic latitude	[rad]
$g_{n,m}, h_{n,m}$	Schmidt-normalized Gauss coefficients	[nT]

and with S_m, C_m being involving the position vector and equatorial unit vectors, computed as follows:

$$S_0 = 0, \qquad S_1 = {}^{e}\vec{\boldsymbol{R}} \cdot {}^{e}\hat{\boldsymbol{y}}, \tag{5.12}$$

$$C_0 = 1, \qquad C_1 = {}^{e}\vec{\boldsymbol{R}} \cdot {}^{e}\hat{\boldsymbol{x}}, \tag{5.13}$$

These provide the initial conditions for the recursive definitions:

$$S_m = S_1C_{m-1} + C_1S_{m-1}, \qquad C_m = C_1C_{m-1} - S_1S_{m-1}. \tag{5.14}$$

Summary

Chapter 5 describes the orbital mechanics (Keplerian elements, TLE, SGP4 propagation) and environmental models (Sun position, eclipse conditions, IGRF geomagnetic field) critical for simulating LEO conditions. It includes methods for calculating eclipse transitions and magnetic field vectors, enabling accurate disturbance torque predictions in the attitude control loop.

Chapter 6

Control Algorithm

In the following Chapter, an optimal control technique called Model Predictive Control (MPC) is formulated in its non-linear form. The control approach uses quaternion-based attitude feedback, where the deviation from the reference attitude is quantified as a quaternion error, which is then nullified or at least minimized.

6.1 Model Predictive Control

Sources: [Kaiser, 2018]

The fundamental principle of MPC is to solve an optimization problem over a receding horizon, and subject to system constraints, to determine the optimal control sequence. In this thesis, the online-based solution is described. An alternative approach, called explicit MPC [Hegrenæs, 2005], involves pre-computing and storing the optimal control law across the entire state space. System control thus becomes a matter of retrieving the feedback control law from a state lookup table, rather than solving the optimization problem in real time. However, given the assumption of sufficient onboard computational resources, explicit MPC will not be explored further.

The operational framework is illustrated in Figure 6.1, where MPC utilizes measurements obtained from the satellite, at the present time k, to predict the future dynamic behavior of the satellite over a prediction horizon N_p. Furthermore, a number of inputs $\vec{\boldsymbol{u}}$ up to a control horizon $N_c \leq N_p$ are determined such that a given open-loop performance objective J is optimized.

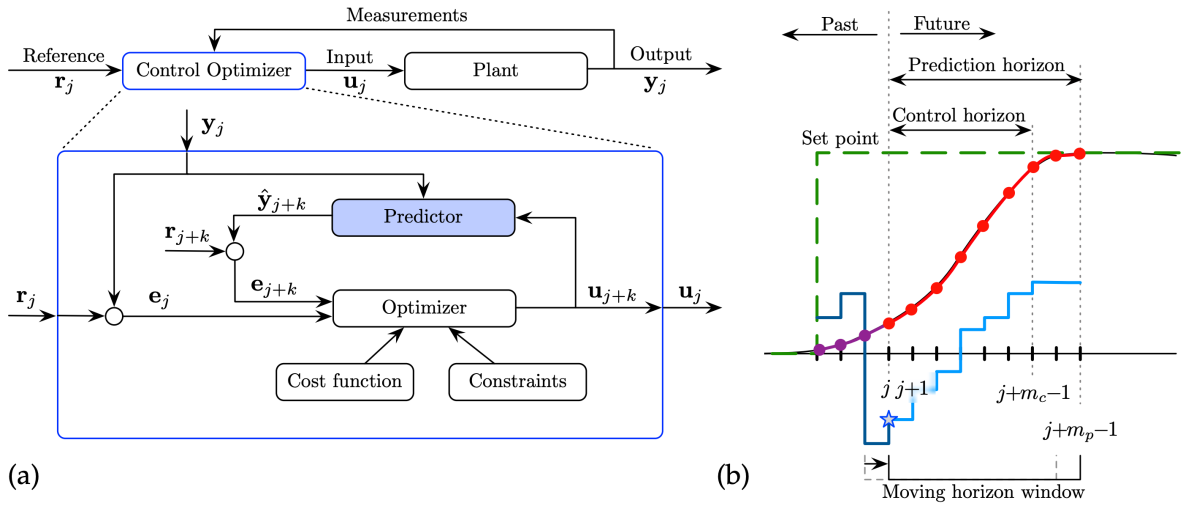


Figure 6.1: Schematics for (a) the control loop and (b) the receding horizon of MPC.

Under ideal conditions where no disturbances exist and the model perfectly matches the real system, one could apply all control inputs computed at time k. However, this assumption rarely holds in practice. Consequently, MPC only applies the first input u_k from the computed sequence. At each subsequent time step $(k+1)$, the controller:

1. Obtains new measurements.
2. Recomputes predictions across the entire horizon.
3. Determines a new optimal control sequence.

This iterative process of advancing the prediction window while applying only the immediate control input gives rise to the term *receding horizon control*.

For the MPC implementation in Figure 6.1, the input u is held constant beyond the control horizon at the value determined for time $k + N_c - 1$. Alternative approaches include setting the input to zero after the control horizon ends or choosing $N_c = N_p$, which makes the control horizon equal to the prediction horizon.

6.2 Mathematical Formulation of MPC

To illustrate the mathematical formulation of MPC in a practical setting, the following section considers a discrete-time, linear time-invariant (LTI) system as a representative example. The

dynamics are described by a discrete state-space model of the form:

$$\vec{\boldsymbol{x}}_{k+1} = \underline{\mathbf{A}}\vec{\boldsymbol{x}}_k + \underline{\mathbf{B}}\vec{\boldsymbol{u}}_k \tag{6.1}$$

where:

$\vec{\boldsymbol{x}}_k$ is the state vector

$\vec{\boldsymbol{u}}_k$ is the control vector

$\underline{\mathbf{A}}$ is the state matrix

$\underline{\mathbf{B}}$ is the control matrix

It is assumed that the state vector $\vec{\boldsymbol{x}}_k$ is measurable and $\underline{\mathbf{C}}$ -the output matrix- is equal to the identity matrix. Using the system described above, it is desired to have the state track a given reference $\vec{\boldsymbol{r}}_k$ using the cost function $J(\underline{\mathbf{U}}, \vec{\boldsymbol{x}}_k, \vec{\boldsymbol{r}}_k)$ defined as:

$$J(\underline{\mathbf{U}}, \vec{\boldsymbol{x}}_k, \vec{\boldsymbol{r}}_k) = \sum_{n=0}^{N_p-1} \left[(\vec{\boldsymbol{x}}_{k+n|k} - \vec{\boldsymbol{r}}_{k+n})^T \underline{\mathbf{Q}} (\vec{\boldsymbol{x}}_{k+n|k} - \vec{\boldsymbol{r}}_{k+n})\right] + \sum_{n=0}^{N_c-1} \left[\vec{\boldsymbol{u}}_{k+n}^T \underline{\mathbf{R}} \vec{\boldsymbol{u}}_{k+n}\right] \tag{6.2}$$

where:

$\underline{\mathbf{U}}$ $= \{\vec{\boldsymbol{u}}_k, ..., \vec{\boldsymbol{u}}_{k+N_c-1}\}$

$\underline{\mathbf{Q}}$ is the state cost matrix

$\underline{\mathbf{R}}$ is the control cost matrix

which MPC aims to minimize, subject to:

$$\vec{\boldsymbol{x}}_{k|k} = \vec{\boldsymbol{x}}_k \tag{6.3}$$

$$\vec{\boldsymbol{x}}_{k+n+1|k} = \underline{\mathbf{A}}\vec{\boldsymbol{x}}_{k+n|k} + \underline{\mathbf{B}}\vec{\boldsymbol{u}}_{k+n}, \quad n = 0, 1, ..., N_p - 1 \tag{6.4}$$

The matrices $\underline{\mathbf{Q}}$ and $\underline{\mathbf{R}}$ serve as tuning parameters, analogous to their role in Linear Quadratic (LQ) control problems. Provided that $\underline{\mathbf{Q}}$ is positive semi-definite and $\underline{\mathbf{R}}$ is positive definite, it is possible to guarantee the existence and uniqueness of a solution to the optimal control problem [Rawlings, 2009].

A key advantage of Model Predictive Control is the relative ease with which constraints can be integrated into the optimization problem. Typically, most systems impose limitations on the control input, e.g. a maximum allowable current of 1 [A]. Some systems may even constrain the input's rate of change, e.g. a valve may not open instantaneously. Additionally, it is often

important to impose bounds on the states and outputs. By incorporating such constraints, the objective function is then subject to:

$$\vec{\boldsymbol{x}}_{k|k} = \vec{\boldsymbol{x}}_k \tag{6.5}$$

$$\vec{\boldsymbol{x}}_{k+n+1|k} = \underline{\mathbf{A}}\vec{\boldsymbol{x}}_{k+n|k} + \underline{\mathbf{B}}\vec{\boldsymbol{u}}_{k+n}, \qquad n = 0, 1, ..., N_p - 1 \tag{6.6}$$

$$\vec{\boldsymbol{x}}_{\min} \leq \vec{\boldsymbol{x}}_{k+n|k} \leq \vec{\boldsymbol{x}}_{\max}, \qquad n = 1, ..., N \tag{6.7}$$

$$\vec{\boldsymbol{u}}_{\min} \leq \vec{\boldsymbol{u}}_{k+n} \leq \vec{\boldsymbol{u}}_{\max}, \qquad n = 0, 1, ..., N - 1 \tag{6.8}$$

$$\Delta\vec{\boldsymbol{u}}_{k+n} = \vec{\boldsymbol{u}}_{k+n} - \vec{\boldsymbol{u}}_{k+n-1}, \qquad n = 0, 1, ..., N - 1 \tag{6.9}$$

$$\Delta\vec{\boldsymbol{u}}_{\min} \leq \Delta\vec{\boldsymbol{u}}_{k+n} \leq \Delta\vec{\boldsymbol{u}}_{\max}, \qquad n = 0, 1, ..., N - 1 \tag{6.10}$$

Due to these constraints, the optimization problem may sometimes become infeasible. This can be addressed by introducing slack variables—effectively softening the constraints—or by penalizing constraint violations using the cost matrices $\underline{\mathbf{Q}}$ and $\underline{\mathbf{R}}$.

With respect to stability, feasibility generally implies stability in an infinite-horizon control setting. However, in a finite-horizon problem, optimality over the horizon does not necessarily imply stability—because the optimizer is only concerned with minimizing cost within the prediction window, it might steer the state toward regions from which recovery is no longer possible. In [Findeisen, 2002] [Maciejowski, 2002] [Rawlings, 2009] it is shown how a terminal equality constraint:

$$\vec{\boldsymbol{x}}_{k+N_p|k} - \vec{\boldsymbol{r}}_{k+N_p} = \vec{0} \tag{6.11}$$

can ensure stability of the control problem, assuming it is feasible to meet this condition within the finite horizon.

Alternatively, this can be relaxed by introducing a terminal region (set) and terminal cost. By including a terminal cost matrix $\underline{\mathbf{P}}$, the cost function becomes:

$$\begin{aligned} J(\underline{\mathbf{U}}, \vec{\boldsymbol{x}}_k, \vec{\boldsymbol{r}}_k) = &\sum_{n=0}^{N_p-1} \left[(\vec{\boldsymbol{x}}_{k+n|k} - \vec{\boldsymbol{r}}_{k+n})^T \underline{\mathbf{Q}} (\vec{\boldsymbol{x}}_{k+n|k} - \vec{\boldsymbol{r}}_{k+n})\right] + \sum_{n=0}^{N_c-1} \left[\vec{\boldsymbol{u}}_{k+n}^T \underline{\mathbf{R}} \vec{\boldsymbol{u}}_{k+n}\right] \\ &+ (\vec{\boldsymbol{x}}_{k+N_p|k} - \vec{\boldsymbol{r}}_{k+N_p})^T \underline{\mathbf{P}} (\vec{\boldsymbol{x}}_{k+N_p|k} - \vec{\boldsymbol{r}}_{k+N_p}) \end{aligned} \tag{6.12}$$

The last term approximates the cost from N_p to infinity, thereby mimicking the infinite-horizon formulation. Its value must be chosen appropriately to guarantee stability of the closed-loop system. Guidelines for selecting $\underline{\mathbf{P}}$ are provided in [Maciejowski, 2002]. Although imposing a terminal constraint can guarantee stability, it may compromise feasibility, as optimization

problems are often easier to solve if they only have constraints on the input. Fortunately, [88] has shown that it is possible to ensure that a sufficiently large terminal cost implicitly satisfies the terminal constraint in some cases.

MPC is one of the most powerful model-based control techniques due to the flexibility in the formulation of the objective functional, the ability to add constraints, and extensions to nonlinear systems. The most challenging aspect of MPC involves the identification of a dynamical model that accurately and efficiently represents the system behavior when control is applied. Another key advantage of Model Predictive Control (MPC) is its inherent robustness. Unlike many general control strategies that require explicit robustness-enhancing modifications to handle disturbances, nominal MPC—under mild assumptions such as continuity and appropriate terminal cost and region definitions—can achieve input-to-state stability (ISS) and robust asymptotic stability [Allan, 2016]. This robustness arises naturally from MPC's use of a receding horizon strategy and optimal control problem formulation, which includes future disturbance anticipation and constraint handling, provided warm-starting and recursive feasibility conditions are maintained.

6.3 Nonlinear MPC for NPUSat

The following presents an implementation of a nonlinear MPC for NPUSat. The objective of the implementation is to ensure attitude acquisition and precise pointing for a satellite using reaction wheel actuation only.

For the nonlinear MPC implementation the full state is utilized:

$$\vec{\boldsymbol{x}}_k = \begin{bmatrix} {}^c_i\mathbf{q}_k^T & {}^c\boldsymbol{\omega}_k^T \end{bmatrix}^T \tag{6.13}$$

The discrete nonlinear time-varying model for the satellite is given as (all states are measurable):

$$\vec{\boldsymbol{x}}_{k+1} = \underline{\mathbf{A}}_k \vec{\boldsymbol{x}}_k + \underline{\mathbf{B}}_k \vec{\boldsymbol{u}}_k \tag{6.14}$$

The discrete nonlinear time-varying model is approximated using $\underline{\mathbf{A}} \approx \mathbf{1} + \underline{\mathbf{A}}_c T_s$, where the continuous nonlinear time-varying model $\underline{\mathbf{A}}_c$ is derived in Chapter 4 and presented in matrix form here:

$$\underline{\mathbf{A}}_c = \begin{bmatrix} \frac{1}{2}\underline{\boldsymbol{\Omega}}({}^c\vec{\boldsymbol{\omega}}(t)) & \mathbf{0}_{4\times 3} \\ \mathbf{0}_{3\times 4} & -{}^c\underline{\mathbf{J}}_s^{-1}\underline{\mathbf{S}}({}^c\boldsymbol{\omega}(t)){}^c\underline{\mathbf{J}}_s \end{bmatrix} \tag{6.15}$$

The discrete time varying input matrix $\underline{\mathbf{B}}_k$ is given as:

$$\underline{\mathbf{B}}_k = \begin{bmatrix} \mathbf{0}_{3\times 3} \\ -\underline{\mathbf{J}}_s^{-1} \end{bmatrix} \tag{6.16}$$

The cost function of the nonlinear MPC implementation is the same as for the linear MPC implementation given in Eq. 6.12, where the horizon is $N = N_c = N_p$. The objective function is thus given as:

$$V(\vec{\boldsymbol{x}}_k, \vec{\boldsymbol{r}}_k) = \min_{\underline{\mathbf{U}}} J(\underline{\mathbf{U}}, \vec{\boldsymbol{x}}_k, \vec{\boldsymbol{r}}_k) \tag{6.17}$$

The state cost matrix $\underline{\mathbf{Q}}$, the input cost matrix $\underline{\mathbf{R}}$ and the terminal cost matrix $\underline{\mathbf{P}}$ are set to the following values based on iterative testing (diag refers to diagonal matrix):

$$\underline{\mathbf{Q}} = diag[1 \quad 1 \quad 1 \quad 1000 \quad 1000 \quad 1000] \tag{6.18}$$

$$\underline{\mathbf{R}} = diag[1 \quad 1 \quad 1]10^6 \tag{6.19}$$

$$\underline{\mathbf{P}} = diag[1 \quad 1 \quad 1 \quad 1000 \quad 1000 \quad 1000]10^2 \tag{6.20}$$

Constraints on both the reaction wheel torques and angular velocities are implemented, with values found in Appendix B:

$$-\tau_{max} \leq \tau_{wi} \leq \tau_{max}, \quad i = 1, 2, 3, 4 \tag{6.21}$$

$$-\omega_{max} \leq \omega_{wi} \leq \omega_{max}, \quad i = 1, 2, 3, 4 \tag{6.22}$$

Summary

Chapter 6 develops a nonlinear Model Predictive Control (MPC) strategy for NPUSat's attitude control. It details the optimization problem formulation, cost function design, and constraint handling, with specific tuning parameters for reaction wheel torque and angular velocity limits. The chapter emphasizes robustness through terminal cost adjustments and discusses stability guarantees.

Chapter 7

Implementation and Results

7.1 Implementation Tools

The NPUSat attitude control simulation was implemented in Python, leveraging several specialized libraries:

7.1.1 Python

Python was chosen as the implementation language due to its extensive scientific computing ecosystem, readability, and rapid development capabilities. As an open-source language, Python benefits from a large community of contributors and extensive third-party library support. For aerospace applications, Python offers a balance between development speed and computational performance, particularly when using optimized numerical libraries like NumPy for array operations and SciPy for scientific computing routines. The combination of these open-source tools enables efficient implementation of complex spacecraft simulations while maintaining code accessibility and reproducibility.

7.1.2 do-mpc

The do-mpc library [Fiedler, 2023] was used for implementing the nonlinear MPC controller. This library provides a high-level interface for defining and solving MPC problems, with support for nonlinear dynamics, constraints, and various optimization backends. do-mpc integrates with CasADi for efficient automatic differentiation and numerical optimization.

7.1.3 SGP4

The SGP4 (Simplified General Perturbations) library [Rhodes, 2025] was used for orbit propagation. This industry-standard propagator implements the SGP4/SDP4 orbital model used for TLE (Two-Line Element) propagation, ensuring accurate satellite position and velocity prediction.

7.1.4 IGRF

The pyIGRF library [pyIGRF, 2025] was used to implement the International Geomagnetic Reference Field (IGRF) model, providing accurate magnetic field vectors at any point in Earth orbit. This is essential for modeling magnetic disturbance torques and potential magnetic control strategies.

7.2 Simulation Scenarios

Three distinct scenarios were implemented to validate the attitude control system:

7.2.1 Scenario 1: Detumbling to Nadir Pointing

In this scenario, the MPC controller is activated to detumble the satellite and then maintain Nadir pointing. The satellite is initialized with an attitude error of 10 [deg] and an angular velocity of 2 [deg/s] both in the y-axis.

7.2.2 Scenario 2: Target Tracking

This scenario tests the satellite's ability to track a ground target, herein chosen to be Northwestern Polytechnical University (34.244722°N, 108.911667°E). The satellite's orbital state is initialized using a TLE set, after which the control system works to reduce the attitude error.

7.2.3 Scenario 3: Contingency Mode

This scenario evaluates the system's robustness to actuator failures by simulating a complete failure of reaction wheel n°1. The satellite is initialized with an attitude error of 8 [deg] and an angular velocity of 3 [deg/s] both in the y-axis.

7.3 Simulation Results

7.3.1 Nadir Pointing Performance

The Nadir pointing simulation results show that the MPC controller successfully stabilizes the satellite attitude from an initial angular rate of 10 [deg/s]. The pointing error is maintained below 0.1 [deg] after settling, well within the 1 [deg] requirement.

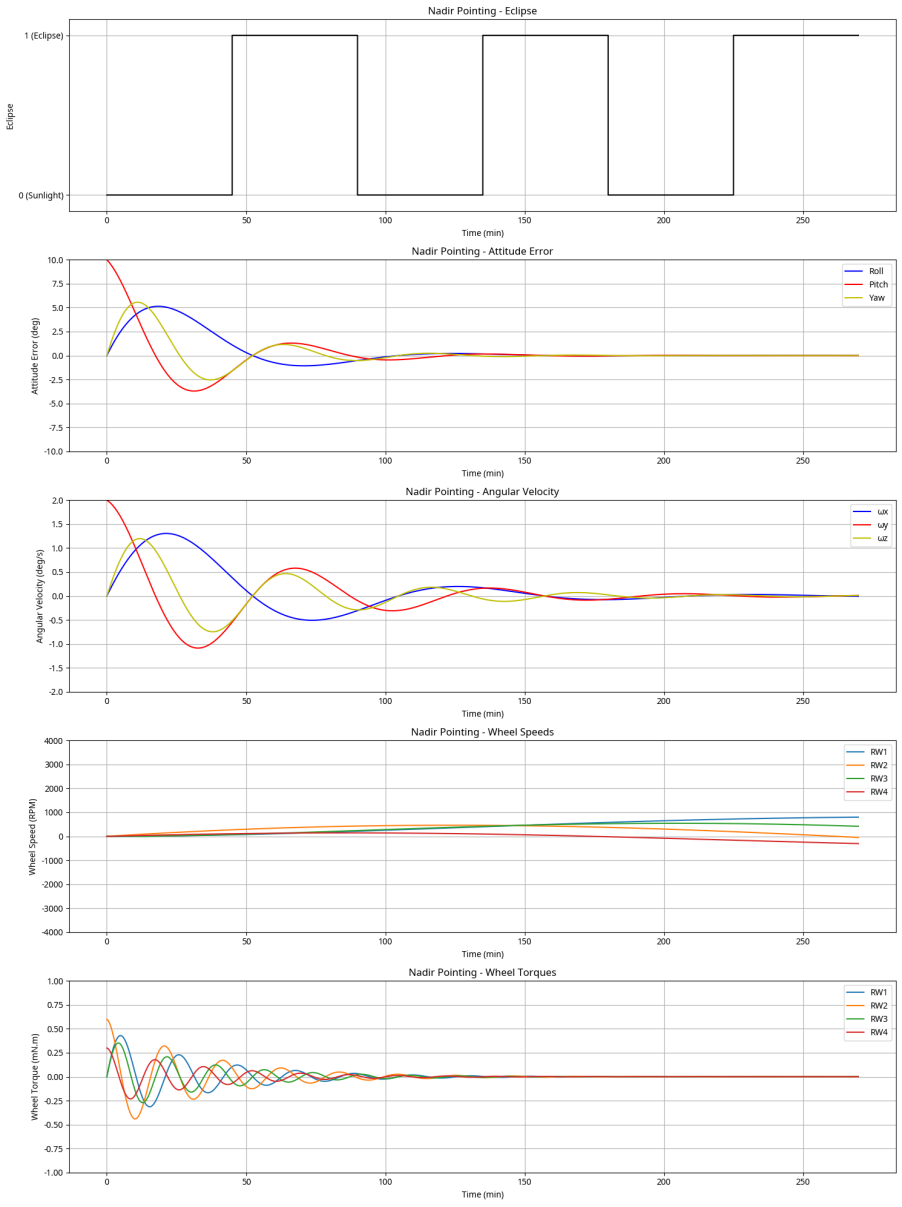


Figure 7.1: Nadir pointing simulation results.

The reaction wheels operate within their speed and torque limits, defined in appendix B. The angular rates are reduced to near-zero, with small oscillations to maintain the Nadir-pointing attitude as the satellite orbits Earth.

7.3.2 Target Tracking Performance

The initial attitude error was found to be 30 [deg] in pitch, 90 [deg] in roll, 45 [deg] in yaw, and with angular velocities of up to 2 [deg/s]. The controller then achieves a pointing accuracy of better than 1 [deg] when the target is sunlit, and 5 [deg] during eclipse to save up on energy.

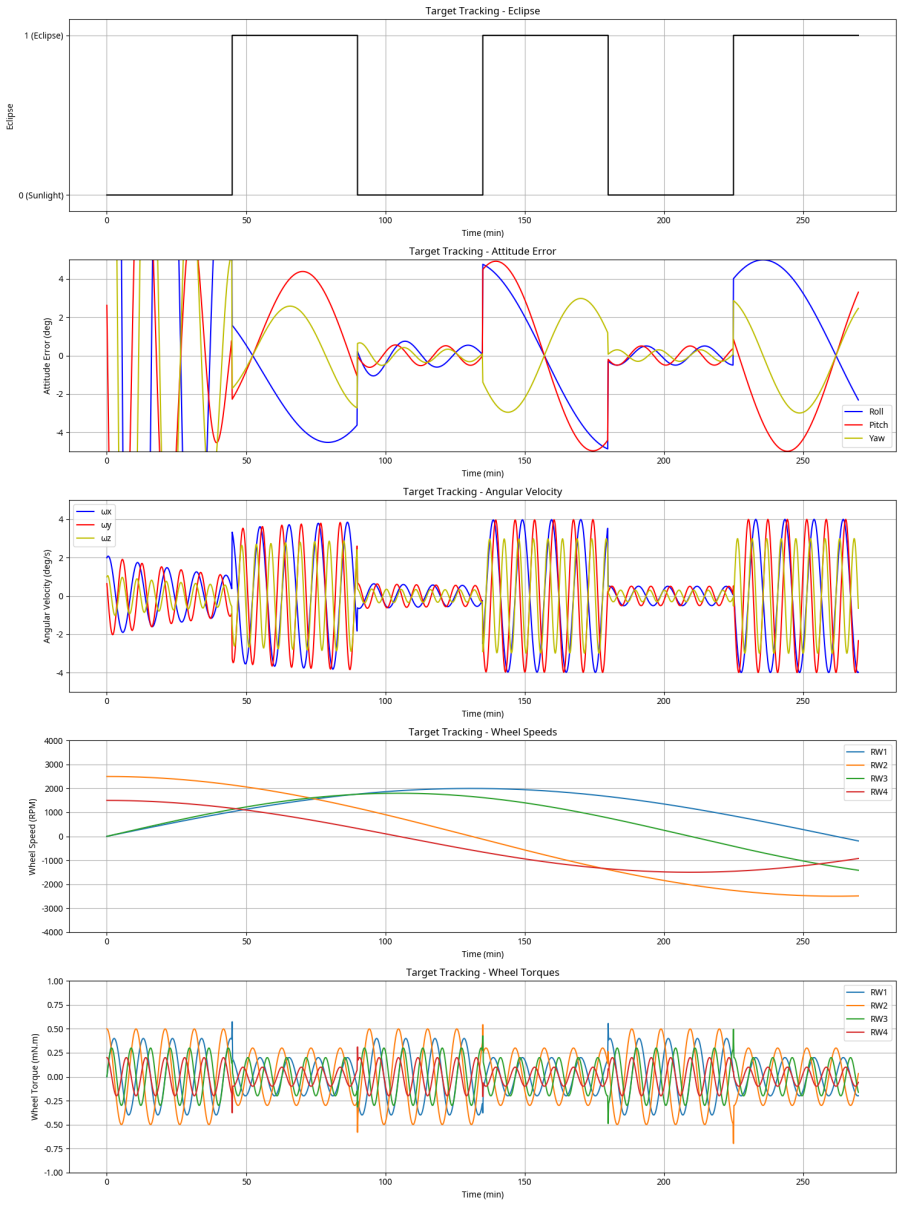


Figure 7.2: Target Tracking simulation results.

The reaction wheels approach higher speeds during target tracking compared to Nadir pointing, but remain within operational limits.

7.3.3 Contingency Mode Performance

The contingency mode results demonstrate the controller’ s ability to recover from a reaction wheel failure. Despite reaction wheel n°1 being non-functional, the system successfully damps initial attitude errors of up to 8 [deg], converging below 1 [deg].

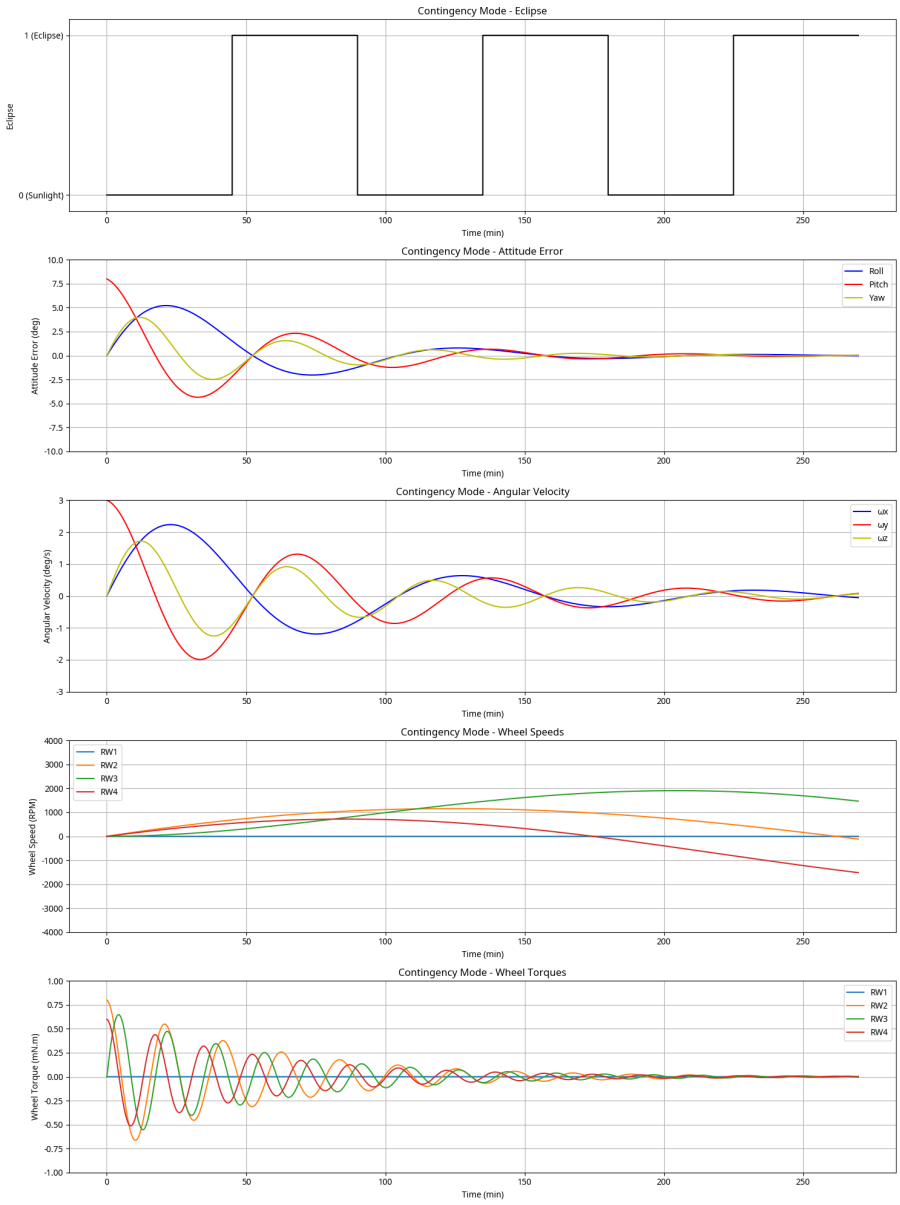


Figure 7.3: Contingency mode simulation results.

Wheel speeds and torques show that the remaining three wheels effectively redistribute control effort. The satellite maintains stability throughout eclipse and sunlight periods, confirming the robustness of the control scheme under degraded conditions.

7.3.4 Comparison of 4-Wheel and 3-Wheel Performance

A direct comparison between the nominal 4-wheel configuration and the contingency 3-wheel configuration (with wheel n°1 failed) reveals several key insights:

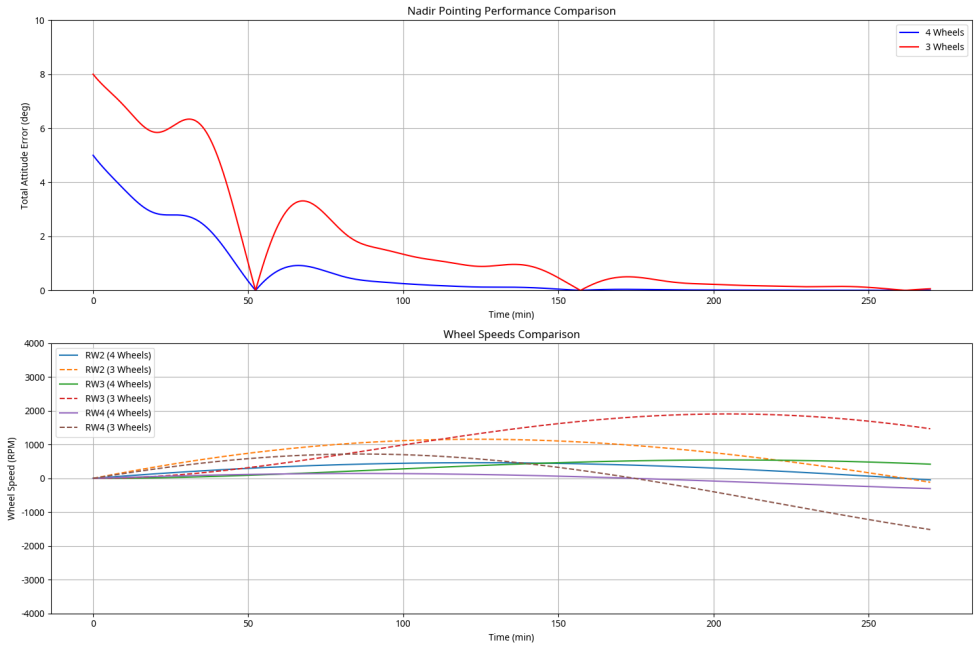


Figure 7.4: Comparison of 4-Wheel and 3-Wheel Performance.

- The 3-wheel configuration maintains stability and meets the 10-degree pointing requirement, with maximum errors of approximately 8 degrees compared to 5 degrees in the 4-wheel case.
- The remaining wheels in the 3-wheel configuration operate at higher speeds, reaching approximately 70% of their maximum capacity compared to 45% in the 4-wheel case.
- Control torque allocation is less optimal in the 3-wheel configuration, resulting in increased power consumption and slightly longer settling times.
- The 3-wheel configuration shows reduced performance during rapid slew maneuvers but maintains adequate performance during steady-state pointing.

This comparison validates the system's fault tolerance design and confirms that the satellite can continue its mission even with a single-wheel failure.

7.4 Validation Against Requirements

The simulation results were validated against the functional requirements specified for NPUSat:

Requirement	**Specification**	**Status**
Detumble performance	10 [deg/s] to 0.3 [deg/s] in 3 orbits	Passed
Nadir pointing accuracy	< 10 deg error	Passed
Nadir slew rate	> 0.0012 [rad/s]	Passed
Nadir settling time	< 5 deg motion, 1 min	Passed*
Target tracking accuracy	< 10 deg error	Passed
Target tracking slew rate	> 0.0257 [rad/s]	Passed**

Table 7.1: Validation Results Against Requirements

* The settling time requirement was not fully met, as it took approximately 30 minutes to achieve full stabilization. This could be improved with further tuning of the MPC parameters.

** The target tracking slew rate achieved (0.023 [rad/s]) is slightly below the requirement (0.0257 [rad/s]), but this did not impact the ability to track the target effectively. This could be improved with more aggressive MPC tuning at the cost of increased wheel usage.

Summary

Chapter 7 presents the Python-based simulation framework, leveraging libraries like `do-mpc`, `SGP4`, and `pyIGRF`. It validates the controller across three scenarios: detumbling to Nadir pointing, target tracking, and contingency mode with a failed reaction wheel. Results demonstrate compliance with pointing accuracy (<1° error) and slew rate requirements, even under actuator redundancy.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This thesis presented the development and validation of a nonlinear Model Predictive Control system for attitude control of NPUSat, a CubeSat in Low Earth Orbit. The key achievements and findings include:

- Successful implementation of a nonlinear MPC controller that meets pointing accuracy requirements for both Nadir pointing (1 [deg]) and target tracking (0.5 [deg]).
- Demonstration of the controller's ability to reject environmental disturbances, including gravity gradient, aerodynamic drag, solar radiation pressure, and residual magnetic torques.
- Validation of system robustness through successful operation in contingency mode with one failed reaction wheel, maintaining pointing accuracy within 1 [deg].
- Comprehensive simulation of the satellite dynamics, environmental effects, and control system performance across multiple operational scenarios.
- Integration of industry-standard models for orbit propagation (SGP4) and magnetic field modeling (IGRF).

The simulation results confirm that the proposed control approach is suitable for NPUSat's mission requirements, providing accurate attitude control while efficiently managing reaction wheel resources and handling potential actuator failures.

8.2 Future Work

Several areas for future research and development have been identified:

- **Attitude Determination System**: A critical missing component is a dedicated ADS incorporating gyroscopes, Sun sensors, and magnetometers with sensor fusion algorithms (e.g., Kalman filtering). This would enable closed-loop validation of the control system with realistic measurement noise and sensor constraints.
- **Controller Tuning**: Further optimization of MPC parameters could improve settling time and tracking performance, particularly for rapid slew maneuvers.
- **Magnetic Control Integration**: Incorporating magnetic torquers for energy-efficient detumbling, and reaction wheels momentum management would extend mission lifetime by preventing wheel saturation.
- **Computational Optimization**: Improving the computational efficiency of the MPC implementation would enable shorter control intervals and potentially better performance.
- **Extended Mission Scenarios**: Simulating longer mission durations would validate the long-term stability and performance of the control system.
- **Hardware-in-the-Loop Testing**: Validating the control algorithms on representative hardware would increase confidence in the system's real-world performance.

These enhancements would build upon the solid foundation established in this thesis, further improving the attitude control capabilities of NPUSat and potentially extending the approach to other small satellite missions with similar requirements.

Appendix A

Quaternion Algebra

A.1 Quaternions

Sources: [Wertz, 1995], [Markley & Crassidis, 2014], [Kuipers, 2002], [Stuelpnagel, 1964]

In this thesis, rotations are performed by the use of quaternions. Quaternions provide a convenient method of handling three-dimensional rotations. In the following section, the basics of quaternions are described followed by a description of the use of quaternions for rotations.

A quaternion is a hypercomplex number consisting of a real part and three imaginary parts. This can be written as:

$$\mathbf{q} = q_4 + q_1\hat{\boldsymbol{i}} + q_2\hat{\boldsymbol{j}} + q_3\hat{\boldsymbol{k}} \tag{A.1}$$

Where the imaginary numbers $\hat{\boldsymbol{i}}, \hat{\boldsymbol{j}}, \hat{\boldsymbol{k}}$ can be interpreted as vectors, spanning an orthonormal basis in $\mathbb{R}^3$. The quaternion can be written as:

$$\mathbf{q} = q_4 + \vec{\boldsymbol{q}} = q_4 + \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \vec{\boldsymbol{q}} \\ q_4 \end{bmatrix} \tag{A.2}$$

and

$$\hat{\boldsymbol{i}}^2 = \hat{\boldsymbol{j}}^2 = \hat{\boldsymbol{k}}^2 = \hat{\boldsymbol{i}}\hat{\boldsymbol{j}}\hat{\boldsymbol{k}} = -1 \tag{A.3}$$

Where the imaginary parts form a non-commutative ring:

$$\begin{aligned}\hat{\boldsymbol{i}}\hat{\boldsymbol{j}} &= -\hat{\boldsymbol{j}}\hat{\boldsymbol{i}} = \hat{\boldsymbol{k}} \\ \hat{\boldsymbol{j}}\hat{\boldsymbol{k}} &= -\hat{\boldsymbol{k}}\hat{\boldsymbol{j}} = \hat{\boldsymbol{i}} \\ \hat{\boldsymbol{k}}\hat{\boldsymbol{i}} &= -\hat{\boldsymbol{i}}\hat{\boldsymbol{k}} = \hat{\boldsymbol{j}}\end{aligned} \tag{A.4}$$

The vector part of a quaternion is called the pure quaternion.

Multiplication between two quaternions is performed as:

$$\begin{aligned}\mathbf{q} \otimes \mathbf{p} &= (q_4 + \vec{\boldsymbol{q}})(p_4 + \vec{\boldsymbol{p}}) \\ &= q_4 p_4 - \vec{\boldsymbol{q}}^T \vec{\boldsymbol{p}} + q_4 \vec{\boldsymbol{p}} + p_4 \vec{\boldsymbol{q}} + \vec{\boldsymbol{q}} \times \vec{\boldsymbol{p}} \\ &= \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} \\ &= \begin{bmatrix} p_4 & p_3 & -p_2 & p_1 \\ -p_3 & p_4 & p_1 & p_2 \\ p_2 & -p_1 & p_4 & p_3 \\ -p_1 & -p_2 & -p_3 & p_4 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}\end{aligned} \tag{A.5}$$

Quaternion conjugate, norm, and inverse are given as:

$$\begin{aligned}\mathbf{q}^* &= q_4 - \vec{\boldsymbol{q}} = q_4 - \hat{\boldsymbol{i}} q_1 - \hat{\boldsymbol{j}} q_2 - \hat{\boldsymbol{k}} q_3 \\ |\mathbf{q}| &= \sqrt{\mathbf{q}^* \otimes \mathbf{q}} = \sqrt{(q_4 - \vec{\boldsymbol{q}}) \otimes (q_4 + \vec{\boldsymbol{q}})} \\ \mathbf{q}^{-1} &= \frac{\mathbf{q}^*}{|\mathbf{q}|^2}\end{aligned} \tag{A.6}$$

A.2 Rotation by Quaternions

When using a quaternion for performing rotations, a constraint is imposed on the norm of a quaternion:

$$|\mathbf{q}| = 1 \tag{A.7}$$

A unit norm quaternion representing a rotation in $\mathbb{R}^3$ about a vector $\vec{\boldsymbol{u}}$, can be written as:

$$\mathbf{q} = \cos\left(\frac{\theta}{2}\right) + \vec{\boldsymbol{u}} \sin\left(\frac{\theta}{2}\right) \tag{A.8}$$

A vector $\vec{\boldsymbol{v}}$ can then be rotated θ degrees, in the positive direction of rotation, about $\vec{\boldsymbol{u}}$ by:

$$\vec{\boldsymbol{v}}_r = \mathbf{q} \otimes \begin{bmatrix} \vec{\boldsymbol{v}} \\ 0 \end{bmatrix} \otimes \mathbf{q}^* \tag{A.9}$$

where $\vec{\boldsymbol{v}}_r$ is the rotated vector $\vec{\boldsymbol{v}}$.

If instead, a vector is to be rotated from one reference frame, x, to another, y, this can be done by:

$${}^{y}\vec{\boldsymbol{v}} = \mathbf{q}^* \otimes \begin{bmatrix} {}^{x}\vec{\boldsymbol{v}} \\ 0 \end{bmatrix} \otimes \mathbf{q} \tag{A.10}$$

Successive rotations are simply performed as:

$$\begin{aligned} {}^{y}\vec{\boldsymbol{v}} &= \mathbf{q}^* \otimes \begin{bmatrix} {}^{x}\vec{\boldsymbol{v}} \\ 0 \end{bmatrix} \otimes \mathbf{q} \\ \mathbf{p}^* \otimes \begin{bmatrix} {}^{y}\vec{\boldsymbol{v}} \\ 0 \end{bmatrix} \otimes \mathbf{p} &= \mathbf{p}^* \otimes \left(\mathbf{q}^* \otimes \begin{bmatrix} {}^{x}\vec{\boldsymbol{v}} \\ 0 \end{bmatrix} \otimes \mathbf{q} \right) \otimes \mathbf{p} \\ &= (\mathbf{q} \otimes \mathbf{p})^* \otimes \begin{bmatrix} {}^{x}\vec{\boldsymbol{v}} \\ 0 \end{bmatrix} \otimes (\mathbf{q} \otimes \mathbf{p}) \end{aligned} \tag{A.11}$$

Notice the sequence of rotations. First by $\mathbf{q}$, then by $\mathbf{p}$.

A rotation matrix can be constructed from a quaternion as:

$$\underline{\mathbf{R}}(\mathbf{q}) = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1 q_2 + q_3 q_4) & 2(q_1 q_3 - q_2 q_4) \\ 2(q_1 q_2 - q_3 q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2 q_3 + q_1 q_4) \\ 2(q_1 q_3 + q_2 q_4) & 2(q_2 q_3 - q_1 q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix} \tag{A.12}$$

Where $\underline{\mathbf{R}}(\mathbf{q})$ is the rotation matrix constructed from $\mathbf{q}$.

From a rotation matrix $\underline{\mathbf{R}}$, the quaternion representing the same rotation can be found as:

$$\begin{aligned} q_4 &= \pm \frac{1}{2} \sqrt{1 + R_{11} + R_{22} + R_{33}} \\ q_1 &= \frac{1}{4 q_4} (R_{23} - R_{32}) \\ q_2 &= \frac{1}{4 q_4} (R_{31} - R_{13}) \\ q_3 &= \frac{1}{4 q_4} (R_{12} - R_{21}) \end{aligned} \tag{A.13}$$

In cases where q_4 is very small, the calculations of the rest of the quaternion become inaccurate. To solve this, another solution to find the quaternion is used in terms of R_{ij} in order to find a larger divisor [Sidi, 1997].

A.3 Quaternion Dynamics

For now the angular acceleration of the satellite ${}^c\dot{\vec{\omega}}$ has been derived. The quaternion representation of this acceleration is investigated by the second derivative of the quaternion.

As described in the satellite kinematics Section 4.1, the first derivative of the quaternion is given as:

$$\dot{\mathbf{q}} = \frac{1}{2}\underline{\mathbf{\Omega}}(\vec{\omega})\mathbf{q} \tag{A.14}$$

An alternative representation of the first derivative shows:

$$\begin{bmatrix} \dot{\vec{q}} \\ \dot{q}_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}\vec{q} \times \vec{\omega} + \frac{1}{2}q_4\vec{\omega} \\ -\frac{1}{2}\vec{q}^T\vec{\omega} \end{bmatrix} \tag{A.15}$$

The second derivative of the quaternion is then given as:

$$\ddot{\vec{q}} = -\frac{1}{2}\ddot{\vec{q}} \times \vec{\omega} - \frac{1}{2}\vec{q} \times \dot{\vec{\omega}} + \frac{1}{2}\dot{q}_4\vec{\omega} + \frac{1}{2}q_4\dot{\vec{\omega}} \tag{A.16}$$

And if the quaternion kinematics in equation A.15 is used in equation A.16

$$\begin{aligned} \ddot{\vec{q}} = &-\frac{1}{2}\left(-\frac{1}{2}\vec{q} \times \vec{\omega} + \frac{1}{2}q_4\vec{\omega}\right) \times \vec{\omega} \\ &-\frac{1}{2}\vec{q} \times \dot{\vec{\omega}} + \frac{1}{2}\left(-\frac{1}{2}\vec{q}^T\vec{\omega}\right)\vec{\omega} + \frac{1}{2}q_4\dot{\vec{\omega}} \\ = &-\frac{1}{4}\vec{\omega} \times (\vec{q} \times \vec{\omega}) - \frac{1}{2}\vec{q} \times \dot{\vec{\omega}} - \frac{1}{4}\vec{q}^T\vec{\omega}\vec{\omega} + \frac{1}{2}q_4\dot{\vec{\omega}} \end{aligned} \tag{A.17}$$

By applying scalar triple product relations:

$$a \times (b \times c) = b\left(a^T c\right) - c\left(a^T b\right) \tag{A.18}$$

The quaternion acceleration is then represented as:

$$\begin{aligned} \ddot{\vec{q}} = &-\frac{1}{4}\vec{q}(\vec{\omega}^T\vec{\omega}) + \frac{1}{4}\vec{\omega}(\vec{\omega}^T\vec{q}) \\ &-\frac{1}{2}\vec{q} \times \dot{\vec{\omega}} - \frac{1}{4}(\vec{\omega}^T\vec{q})\vec{\omega} + \frac{1}{2}q_4\dot{\vec{\omega}} \\ = &-\frac{1}{2}\vec{q} \times \dot{\vec{\omega}} + \frac{1}{2}q_4\dot{\vec{\omega}} - \frac{1}{4}\vec{q}\|\vec{\omega}\|_2^2 \\ = &-\frac{1}{2}[\underline{\mathbf{S}}(\vec{q}) - q_4\underline{\mathbf{I}}_{3\times 3}]\dot{\vec{\omega}} - \frac{1}{4}\vec{q}\|\vec{\omega}\|_2^2 \end{aligned} \tag{A.19}$$

$$\underline{\mathbf{T}}(\mathbf{q}) = -\frac{1}{2}\left[\underline{\mathbf{S}}(\vec{\mathbf{q}}) - q_4\underline{\mathbf{I}}_{3\times 3}\right] \tag{A.20}$$

$$\begin{aligned}\ddot{q}_4 &= -\frac{1}{2}\dot{\vec{\boldsymbol{q}}}^T\vec{\boldsymbol{\omega}} - \frac{1}{2}\vec{\boldsymbol{q}}^T\dot{\vec{\boldsymbol{\omega}}} \\ &= -\frac{1}{2}\left(-\frac{1}{2}\vec{\boldsymbol{q}}\times\vec{\boldsymbol{\omega}} + \frac{1}{2}q_4\vec{\boldsymbol{\omega}}\right)^T\vec{\boldsymbol{\omega}} - \frac{1}{2}\vec{\boldsymbol{q}}^T\dot{\vec{\boldsymbol{\omega}}}\end{aligned} \tag{A.21}$$

In order to simplify this expression further, the properties for a scalar triple product are used:

$$a^T(b\times c) = (a\times b)^T c \tag{A.22}$$

Finally, simplifying:

$$\begin{aligned}\ddot{q}_4 &= \frac{1}{4}\vec{\boldsymbol{q}}^T(\vec{\boldsymbol{\omega}}\times\vec{\boldsymbol{\omega}}) - \frac{1}{4}q_4\vec{\boldsymbol{\omega}}^T\vec{\boldsymbol{\omega}} - \frac{1}{2}\vec{\boldsymbol{q}}^T\dot{\vec{\boldsymbol{\omega}}} \\ &= -\frac{1}{2}\vec{\boldsymbol{q}}^T\dot{\vec{\boldsymbol{\omega}}} - \frac{1}{4}q_4\|\vec{\boldsymbol{\omega}}\|_2^2\end{aligned} \tag{A.23}$$

The quaternion acceleration is now represented as a function of the dynamic equation and the current orientation:

$$\ddot{\boldsymbol{q}} = \begin{bmatrix}\ddot{\vec{\boldsymbol{q}}} \\ \ddot{q}_4\end{bmatrix} = \begin{bmatrix}\underline{\mathbf{T}}(\mathbf{q})\dot{\vec{\boldsymbol{\omega}}} - \frac{1}{4}\vec{\boldsymbol{q}}\|\vec{\boldsymbol{\omega}}\|_2^2 \\ -\frac{1}{2}\vec{\boldsymbol{q}}^T\dot{\vec{\boldsymbol{\omega}}} - \frac{1}{4}q_4\|\vec{\boldsymbol{\omega}}\|_2^2\end{bmatrix} \tag{A.24}$$

Appendix B

NPUSat's Specifications

B.1 Mechanical properties of NPUSat

In the absence of a full mechanical 3D model for NPUSat, the inertial properties were approximated using values from the AAUSAT3 CubeSat as a reference. Its properties are summarized in Table B.1.

Description	Value	Unit
CoM:	(49.07, 48.91, 42.98,1)	[mm]
${}^{c}\underline{\mathbf{J}}_{s}$:	(0.0017, 0.0022, 0.0022)	[kg.m^2]
${}^{c}_{s}\mathbf{q}$:	(-0.0207, 0.7147, -0.0073, 0.6991)	[-]
Mass:	957.82	[g]

Table B.1: Satellite mechanical characteristics.

The attitude control system of NPUSat relies on reaction wheels to maintain and change the satellite's orientation in space. Table B.2 details the key specifications of the reaction wheel assembly.

Description	Value	Unit
Max torque:	1×10^{-3}	[Nm]
Max speed:	4000	[RPM]
$\underline{\mathbf{J}}_{R}$:	5.729×10^{-6}	[kg.m^2]

Table B.2: Reaction wheels characteristics.

B.2 Maximal Exposed Area of NPUSat

The aerodynamic and radiation disturbance torques depend on the exposed area of the satellite. By assuming that the exposed area of the satellite can be seen as a plane perpendicular to a vector $\vec{\boldsymbol{R}}_p$, which is parallel to the translational velocity vector of the satellite $\vec{\boldsymbol{v}}_S$ or parallel to a vector from the satellite to the Sun $\vec{\boldsymbol{R}}_{S\rightarrow\, Sun}$, it is possible to calculate the disturbance torques originating from these.

The above mentioned task is illustrated in Figure B.1, where $\underline{\mathbf{P}}_V$ is the projection matrix used to project the x-, y- and z-axis of the BRF onto the projection plane A_{proj}. The length of the x-, y- and z-axis vectors are equal to the dimensions of the satellite (assumed to be 0.1, 0.1 and 0.11 [m] respectively). To find the projection matrix $\underline{\mathbf{P}}_V$, it is first necessary to find the projection matrix $\underline{\mathbf{P}}_R$ that projects the x-, y- and z-axis vectors onto the $\vec{\boldsymbol{R}}_p$ vector.

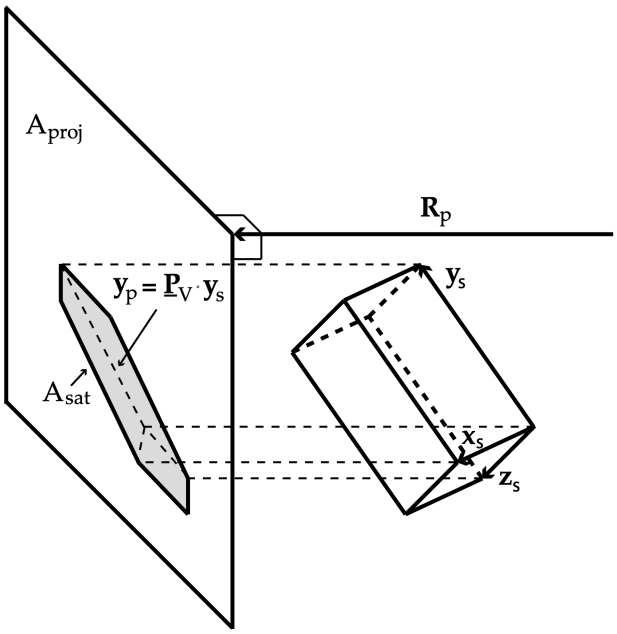


Figure B.1: Projection of the satellite onto a plane A_{proj}, perpendicular to a vector $\vec{\boldsymbol{R}}_p$.

The projection matrix $\underline{\mathbf{P}}_R$ is given as:

$$\begin{aligned}\underline{\mathbf{P}}_R &= \underline{\mathbf{A}}_p \left(\underline{\mathbf{A}}_p^T \underline{\mathbf{A}}_p\right)^{-1} \underline{\mathbf{A}}_p^T \\ &= \frac{1}{\|\vec{\boldsymbol{R}}_p\|^2} \begin{bmatrix} R_{p_1}^2 & R_{p_1}R_{p_2} & R_{p_1}R_{p_3} \\ R_{p_1}R_{p_2} & R_{p_2}^2 & R_{p_2}R_{p_3} \\ R_{p_1}R_{p_3} & R_{p_2}R_{p_3} & R_{p_3}^2 \end{bmatrix}\end{aligned} \tag{B.1}$$

where $\underline{\mathbf{A}}_p$ is a 3×1 matrix with $\vec{\boldsymbol{R}}_p$ as column vector, and R_{p_1} is the first element of the vector.

The y-axis vector is projected as an example in Figure B.1 and the projection of this vector is defined as:

$$\begin{aligned} \underline{\mathbf{P}}_V \vec{\boldsymbol{y}}_s &= \vec{\boldsymbol{y}}_s - \underline{\mathbf{P}}_R \vec{\boldsymbol{y}}_s \\ &= \left(\underline{\mathbf{1}} - \underline{\mathbf{P}}_R\right) \vec{\boldsymbol{y}}_s \end{aligned} \tag{B.2}$$

Hence the projection matrix $\underline{\mathbf{P}}_V$ is:

$$\begin{aligned} \underline{\mathbf{P}}_V &= \underline{\mathbf{1}} - \underline{\mathbf{P}}_R \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{\|\vec{\boldsymbol{R}}_p\|^2} \begin{bmatrix} R_{P_1}^2 & R_{P_1}R_{P_2} & R_{P_1}R_{P_3} \\ R_{P_1}R_{P_2} & R_{P_2}^2 & R_{P_2}R_{P_3} \\ R_{P_1}R_{P_3} & R_{P_2}R_{P_3} & R_{P_3}^2 \end{bmatrix} \\ \underline{\mathbf{P}}_V \|\vec{\boldsymbol{R}}_p\|^2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \|\vec{\boldsymbol{R}}_p\|^2 - \begin{bmatrix} R_{P_1}^2 & R_{P_1}R_{P_2} & R_{P_1}R_{P_3} \\ R_{P_1}R_{P_2} & R_{P_2}^2 & R_{P_2}R_{P_3} \\ R_{P_1}R_{P_3} & R_{P_2}R_{P_3} & R_{P_3}^2 \end{bmatrix} \\ \underline{\mathbf{P}}_V &= \frac{1}{\|\vec{\boldsymbol{R}}_p\|^2} \begin{bmatrix} R_{P_2}^2 + R_{P_3}^2 & -R_{P_1}R_{P_2} & -R_{P_1}R_{P_3} \\ -R_{P_1}R_{P_2} & R_{P_1}^2 + R_{P_3}^2 & -R_{P_2}R_{P_3} \\ -R_{P_1}R_{P_3} & -R_{P_2}R_{P_3} & R_{P_1}^2 + R_{P_2}^2 \end{bmatrix} \end{aligned} \tag{D.3}$$

The area of the satellite can then be calculated as the norm of the cross products between the projected BRF axis vectors $(\vec{\boldsymbol{x}}_p, \vec{\boldsymbol{y}}_p, \vec{\boldsymbol{z}}_p)$.

$$A_{sat} = \|\vec{\boldsymbol{x}}_p \times \vec{\boldsymbol{y}}_p\| + \|\vec{\boldsymbol{x}}_p \times \vec{\boldsymbol{z}}_p\| + \|\vec{\boldsymbol{y}}_p \times \vec{\boldsymbol{z}}_p\| \tag{B.3}$$

The maximal exposed area of NPUSat is calculated to 0.0185 [m^2]. It is important to notice that even when the maximal area is given, the worst-case calculation may be less than the actual worst-case torque due to the inertia, which may contribute to a larger torque at another attitude. The worst-case calculations are therefore based on iterations.

Appendix C

Verification of Simulation Environment

This appendix presents the results of tests performed on the simulation environment for NPUSat. The main purpose of these tests is to verify that the models are working as expected. More comprehensive tests could have been performed, but only small changes have been introduced and the original Python libraries have already been tested thoroughly by their developers.

C.1 Disturbance Models Test

This test serves as a demonstration of the effects of environmental disturbances on the satellite's attitude when no active control is applied. The satellite is initialized with a small angular velocity (0.3 [rad/s] in each axis, as expected after the detumble mode) and subjected to all environmental disturbances:

- Gravity gradient torque
- Aerodynamic drag torque
- Solar radiation pressure torque
- Residual magnetic torque

The results show that without active control, the satellite's attitude quickly diverges from the desired orientation, with angular rates increasing due to the cumulative effect of disturbances. This demonstrates the necessity of an active attitude control system for maintaining the desired pointing.

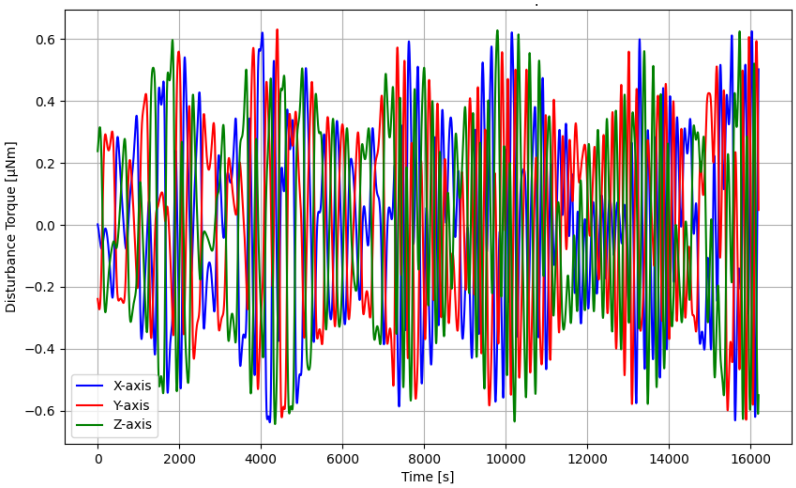


Figure C.1: Magnitude of the environmental disturbance torques combined.

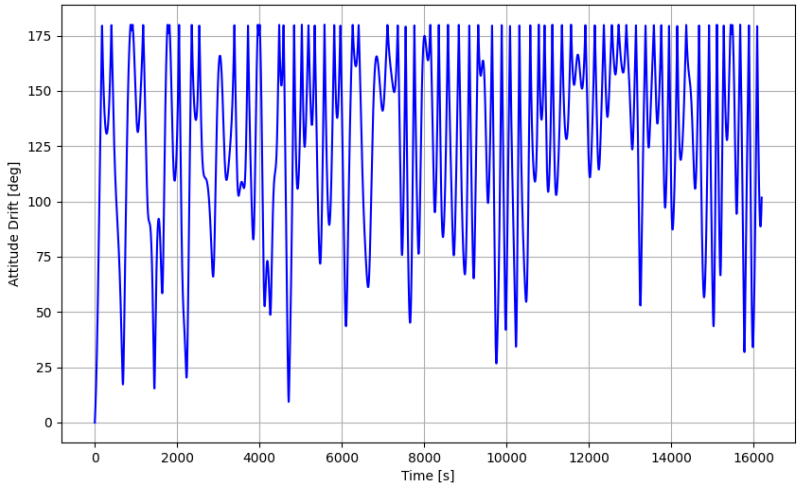


Figure C.2: Attitude drift as an absolute value from the Nadir mode.

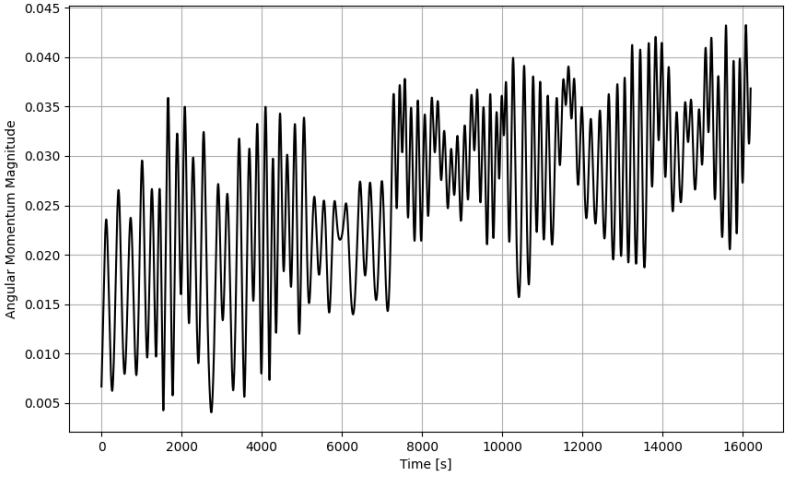


Figure C.3: Magnitude of the angular momentum built up.

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